

**A STUDY ON ATLIQ MART'S
PROMOTIONAL EFFECTIVENESS
DURING DIWALI AND SANKRANTI
CAMPAIGNS**

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SYNOPSIS

In this project, we embarked on a detailed exploration of AtliQ Mart's retail landscape, deploying advanced analytics methodologies and cutting-edge data visualization tools to unearth actionable insights. Our journey commenced with a meticulous examination of sales trends across various product categories, identifying patterns and outliers to inform strategic decision-making. Leveraging key performance indicators such as Incremental Sold Units and Incremental Revenue, we scrutinized the efficacy of promotional campaigns, dissecting their impact on overall sales performance and customer engagement. Furthermore, our analysis extended to assessing store performance metrics and product optimization strategies, aiming to maximize operational efficiency and drive revenue growth. By scrutinizing the City Performance Index and delving into average order values before and after promotions, we uncovered invaluable opportunities for enhancing AtliQ Mart's competitive edge and fostering sustainable business growth. Through our data-driven approach, we not only provided actionable recommendations for immediate implementation but also laid the groundwork for ongoing strategic refinement and innovation within AtliQ Mart's retail ecosystem.

CHAPTER I

INTRODUCTION

The retail landscape, particularly in the Fast-Moving Consumer Goods (FMCG) sector, is witnessing a significant paradigm shift with the increasing adoption of advanced analytics. AtliQ Mart, a prominent player in this dynamic industry, recognizes the transformative potential of data-driven decision-making. Operating a network of 50 supermarkets across the southern region of India, AtliQ Mart is strategically positioned to harness the power of retail analytics for optimizing sales strategies and promotional initiatives. In recent years, the retail sector has evolved from traditional models to ones heavily reliant on data analytics for strategic insights. AtliQ Mart's commitment to leveraging these insights aligns with the broader industry trend where companies seek to gain a competitive edge through a deeper understanding of consumer behaviour and market dynamics.

Within the broader context of retail analytics, this study delves into the intricate dynamics of sales and promotions. These two components play a pivotal role in shaping the success of retail businesses. Sales and promotions not only drive revenue but also serve as essential tools for fostering customer loyalty and positioning a brand in a crowded market.

Sales, as a fundamental metric, encapsulates the overall performance of a retail business. However, promotions add a layer of complexity, as their effectiveness goes beyond immediate sales figures. A well-executed promotion can influence customer perceptions, enhance brand visibility, and set the stage for long-term profitability. This research focuses specifically on dissecting the dynamics of promotional campaigns conducted during Diwali 2023 and Sankranti 2024 at AtliQ Mart.

The contemporary retail landscape presents a myriad of challenges for businesses, especially in the FMCG sector. AtliQ Mart, in its pursuit of sustained growth, faces specific challenges related to optimizing promotional activities. Understanding customer behaviour, aligning product offerings with market trends, and ensuring promotional campaigns resonate with the target audience are complex tasks that demand nuanced decision-making.

The need for this study arises from the recognition that data-driven insights can be a catalyst in overcoming these challenges. By thoroughly analysing the performance of past promotions, the study aims to unravel actionable insights that address existing decision-making gaps. The sales director at AtliQ Mart can then leverage these insights to refine strategic planning, enhance resource allocation, and elevate the overall efficacy of future promotional campaigns.

In undertaking a comprehensive analysis, this study taps into key data sources within AtliQ Mart's ecosystem. The 'dim_campaigns,' 'dim_products,' 'dim_stores,' and 'fact_events' tables form the foundation for extracting meaningful insights. The temporal scope of the study is defined by the Diwali 2023 and Sankranti 2024 promotional periods, strategically chosen due to the unique market conditions prevalent during these festive seasons.

Geographically, the study focuses on the southern region of India, where AtliQ Mart has strategically positioned its 50 supermarkets. This geographical focus ensures a targeted exploration of market conditions and customer behaviours specific to AtliQ Mart's regional operations. By narrowing the scope, the study aims for a more in-depth understanding, providing granular insights that are directly applicable to the market dynamics in which AtliQ Mart operates.

CHAPTER 2

OBJECTIVES OF THE STUDY

2.1 Primary Objective

The primary objective of this study is to optimize promotional strategies by analyzing and enhancing the effectiveness of the Diwali 2023 and Sankranti 2024 promotional campaigns conducted by AtliQ Mart across its 50 supermarkets in the southern region of India. By evaluating the performance of these promotions, the study aims to identify successful strategies and areas for improvement, enabling AtliQ Mart to refine its promotional tactics for sustained growth and competitiveness in the retail market.

2.2 Secondary Objectives

Understand Customer Behavior: This study seeks to delve into customer behavior during promotional periods, aiming to uncover insights that can inform product offerings and future campaign strategies. By analyzing customer purchase patterns, preferences, and responses to promotions, the study aims to gain a deeper understanding of consumer behavior, enabling AtliQ Mart to tailor its offerings to meet customer needs and preferences effectively.

Enhance Decision-Making: Through data-driven insights derived from the analysis, the study aims to address decision-making gaps within AtliQ Mart's promotional strategies. By providing actionable insights into the performance of different promotional activities, the study aims to empower decision-makers to make informed choices regarding resource allocation, campaign optimization, and strategic planning for future promotional initiatives.

Analyze Sales Performance: A key aspect of this study is to evaluate the overall sales performance of AtliQ Mart during the Diwali 2023 and Sankranti 2024 campaigns. By

analyzing sales data and key performance indicators such as Incremental Sold Units, Incremental Revenue, and Average Order Value, the study aims to assess the effectiveness of promotional activities and identify opportunities for improving sales performance and business growth.

Study Regional Dynamics: In addition to analyzing sales performance and promotional effectiveness, the study aims to gain insights into regional dynamics within the southern region of India. By examining market conditions, customer behaviors, and regional preferences, the study aims to provide AtliQ Mart with actionable data to optimize its operations, tailor its offerings, and effectively target regional markets for enhanced business success.

CHAPTER 3

REVIEW OF LITERATURE

Retail analytics, driven by the surge in data availability and advancements in technology, has become an indispensable tool for businesses in the Fast-Moving Consumer Goods (FMCG) sector. This literature review explores the evolving landscape of retail analytics, with a specific focus on understanding its impact on promotional effectiveness within the FMCG domain. The review aims to synthesize existing knowledge, identify research gaps, and set the stage for the current study on the Diwali 2023 and Sankranti 2024 promotions at AtliQ Mart.

The retail industry has witnessed a paradigm shift in recent years, transitioning from traditional approaches to data-centric decision-making. Scholars (Smith et al., 2018; Chen et al., 2020) highlight the significance of retail analytics in enhancing operational efficiency, customer engagement, and overall business performance. The integration of advanced analytics tools, such as machine learning and predictive modeling, has empowered retailers to glean actionable insights from vast datasets.

A substantial body of literature delves into the complex dynamics of promotional strategies within the FMCG sector. Researchers (Jones & Brown, 2017; Gupta et al., 2019) emphasize the critical role promotions play in driving sales, influencing consumer behavior, and shaping brand perceptions. However, the effectiveness of promotions is contingent on various factors, including product characteristics, pricing structures, and market conditions.

Studies (Li et al., 2016; Wang et al., 2021) underscore the synergistic relationship between retail analytics and promotional optimization. Advanced analytics tools enable retailers to analyze historical sales data, identify patterns, and tailor promotions to align

with consumer preferences. The ability to forecast demand, optimize pricing strategies, and personalize promotional campaigns contributes to improved promotional effectiveness.

Despite the advancements, scholars (Chen & Wang, 2018; Lee & Kim, 2020) acknowledge challenges in the application of retail analytics, such as data privacy concerns, implementation costs, and the need for skilled personnel. However, these challenges present opportunities for further research and innovation, particularly in developing ethical frameworks, cost-effective solutions, and training programs.

While existing literature provides valuable insights, there is a notable gap concerning the specific analysis of promotional effectiveness during festive periods in the FMCG sector, as evidenced by the limited studies exploring the impact of promotions during cultural festivals like Diwali and Sankranti. The synthesis of existing knowledge underscores the need for nuanced analyses tailored to specific contexts, such as festive promotions. By addressing these research gaps, the current study aims to contribute to the growing body of literature on retail analytics and enhance our understanding of promotional effectiveness within the dynamic landscape of FMCG retail.

CHAPTER 4

RESEARCH METHODOLOGY

The research aims to analyze the effectiveness of promotional campaigns conducted by AtliQ Mart during the festive periods of Diwali 2023 and Sankranti 2024 across 50 supermarkets in Southern India. The study seeks to evaluate the success levels of these promotions and identify key insights to inform future marketing strategies. By examining sales transactions, promotional activities, and customer behavior, the research endeavors to optimize promotional strategies, enhance decision-making, and improve overall business performance for AtliQ Mart.

4.1 Data Collection

The research relied on comprehensive data collected from Code Basics Resume Challenge "Analyse Promotions and Provide Tangible Insights to Sales Director", comprising sales transactions, promotional activities, product information, and store locations. The dataset covers the entire duration of the Diwali 2023 and Sankranti 2024 promotional campaigns across all 50 supermarkets in Southern India. The data collection process ensured the inclusion of all relevant information necessary for the analysis, allowing for a thorough examination of the effectiveness of promotional activities and their impact on sales performance.

4.2 Data Preparation

Prior to analysis, the collected data underwent a rigorous preparation process to ensure its accuracy, consistency, and suitability for analysis. This involved data cleaning to identify and rectify any inconsistencies, errors, or missing values within the dataset. Additionally, data transformation techniques were employed to standardize formats, reconcile discrepancies, and integrate disparate data sources into a unified dataset. By

preparing the data in this manner, the research ensured the integrity and reliability of the dataset, laying the foundation for meaningful analysis and interpretation.

4.3 Data Analysis

Various statistical and analytical techniques were employed to analyze the dataset and derive actionable insights. Key performance indicators (KPIs) such as Incremental Sold Units, Incremental Revenue, and ISU Percentage were calculated to assess the effectiveness of promotional campaigns and their impact on sales performance. Correlation analysis was conducted to explore relationships between variables such as product category, promotion type, and campaign effectiveness. The analysis aimed to uncover patterns, trends, and relationships within the data, providing valuable insights to inform decision-making and strategy development.

4.4 Tools and Techniques

The analysis for this project was facilitated using a combination of robust tools and techniques, each selected for its unique capabilities and the value it brings to the overall analysis. The primary tools and techniques utilized include SQL for data querying and manipulation and Power BI for data visualization. Here's an elaboration on why each of these tools was chosen and how they contribute to the project's objectives.

SQL for Data Querying and Manipulation

SQL (Structured Query Language) is an essential tool in the domain of data management and analysis, widely regarded as the standard language for interacting with relational databases. It is a highly versatile language that allows users to define, manipulate, and query data, making it an indispensable asset for data analysts, database administrators, and developers. SQL's ability to handle large volumes of data with high efficiency and precision makes it particularly valuable in enterprise environments

where data integrity and accuracy are paramount. One of the key strengths of SQL is its capability to perform complex queries, enabling users to retrieve specific data points from extensive datasets. This includes filtering, sorting, and aggregating data to meet various analytical needs. SQL's robust set of functions and operators allows for the transformation of raw data into meaningful insights, facilitating detailed data analysis. Moreover, SQL supports a variety of data types and complex joins, making it possible to combine data from multiple tables, which is crucial for comprehensive data analysis.

SQL also excels in data manipulation, allowing users to insert, update, delete, and modify data seamlessly. This feature is critical for maintaining up-to-date and accurate databases. Additionally, SQL's transaction control commands ensure data consistency and integrity, even in multi-user environments, by providing mechanisms to manage concurrent access and recover from failures. Another significant advantage of SQL is its standardized nature, which ensures compatibility across different database management systems (DBMS) such as MySQL, PostgreSQL, Microsoft SQL Server, and Oracle. This standardization facilitates the transfer of skills and knowledge across various platforms, making SQL a universal tool in the data management landscape. Furthermore, SQL's integration capabilities with various programming languages and data processing tools enhance its utility, allowing for the creation of sophisticated data workflows and applications.

- **Data Aggregation and Organization:** SQL is highly efficient for aggregating large datasets, performing complex queries, and organizing data in a structured format. It allows for the extraction of specific data subsets required for detailed analysis.
- **Versatility and Precision:** With SQL, it is possible to execute precise queries

that filter, join, and aggregate data from multiple tables. This versatility ensures that the data is prepared accurately, with all necessary transformations applied before further analysis.

- **Foundation for Analysis:** The structured data obtained through SQL queries serves as the foundational dataset for subsequent analyses, ensuring that the data fed into visualization tools is clean and well-organized.

Power BI for Data Visualization

Power BI, developed by Microsoft, is a leading business intelligence tool that empowers organizations to transform their raw data into meaningful insights through interactive and visually compelling dashboards. It is designed to cater to the needs of business analysts, data scientists, and decision-makers, providing a platform that simplifies the process of data visualization and analysis. Power BI's user-friendly interface and robust features make it a powerful tool for uncovering trends, identifying patterns, and making data-driven decisions. One of Power BI's standout features is its ability to integrate seamlessly with a wide array of data sources, including databases, spreadsheets, cloud services, and on-premises data systems. This flexibility ensures that users can consolidate data from multiple sources into a single, coherent view, facilitating comprehensive analysis. Power BI supports real-time data connections, enabling the creation of dashboards that reflect the most current data, which is crucial for timely decision-making.

The platform's advanced visualization capabilities allow users to create a variety of charts, graphs, and maps, making it easier to interpret complex data sets. These visualizations are not only aesthetically pleasing but also interactive, allowing users to drill down into specific data points, apply filters, and view the data from different

perspectives. This interactivity helps stakeholders to engage with the data more effectively and gain deeper insights. Power BI also offers powerful data modeling tools, enabling users to create relationships between different data sets and perform complex calculations using the Data Analysis Expressions (DAX) language. This feature enhances the analytical depth of the reports and dashboards, providing more granular insights. Additionally, Power BI's built-in AI and machine learning capabilities can help identify trends and anomalies in the data, offering predictive insights that can guide strategic planning.

Another significant advantage of Power BI is its collaborative features, which allow users to share reports and dashboards with team members and stakeholders easily. The platform supports role-based access controls, ensuring that sensitive data is protected while still enabling broad access to necessary insights. Power BI's cloud-based service ensures that users can access their dashboards from anywhere, facilitating remote work and collaboration. The benefits of using Power BI include:

- **Interactive Dashboards:** Power BI enables the creation of interactive dashboards that allow stakeholders to explore data dynamically. Users can drill down into specific areas of interest, filter data on the fly, and view data from multiple perspectives.
- **Enhanced Data Understanding:** Visualizations help in making complex data more understandable. Charts, graphs, and other visual aids can highlight trends, patterns, and anomalies that might not be immediately evident in raw data tables.
- **Real-Time Insights:** Power BI supports real-time data updates, which means stakeholders can access the most current data and insights, facilitating timely

decision-making.

Integration of Tools and Techniques

Integrating SQL and Power BI represents a powerful combination that significantly enhances the capability to manage, analyze, and visualize data. This integration leverages the strengths of both tools, enabling organizations to harness the full potential of their data. SQL's robust querying and data manipulation capabilities, coupled with Power BI's advanced data visualization and analytical features, create a comprehensive data analysis ecosystem. The integration begins with SQL, where data is stored in structured relational databases. SQL serves as the foundational layer for data management, enabling users to efficiently extract, transform, and load (ETL) data. SQL queries are used to aggregate, filter, and organize data, ensuring that only the relevant data is prepared for further analysis. This step is crucial for creating a clean and well-structured dataset, which is the backbone of any meaningful data analysis.

Once the data is prepared in SQL, it is then connected to Power BI for visualization and deeper analysis. Power BI provides seamless connectivity to various SQL databases, including MySQL, PostgreSQL, Microsoft SQL Server, and Oracle, among others. This integration allows Power BI to pull data directly from SQL databases, ensuring that the data is always up-to-date and reflecting the latest changes. In Power BI, the imported SQL data can be transformed using Power Query, which offers a range of data transformation tools. Users can merge, append, and clean data, ensuring that it meets the specific requirements of their analysis. Power Query's intuitive interface simplifies these tasks, even for users with limited technical expertise. The true power of this integration is realized in Power BI's data modeling and visualization capabilities. Data from SQL can be modeled using relationships and calculated fields, enhancing

the analytical possibilities. The Data Analysis Expressions (DAX) language in Power BI enables complex calculations and aggregations, allowing for sophisticated data analysis.

Visualizations in Power BI provide a dynamic and interactive way to explore SQL data. Users can create a variety of charts, graphs, and maps to represent the data visually. These visualizations are not only visually appealing but also interactive, allowing users to drill down into specific data points, apply filters, and see real-time updates. This interactivity helps in uncovering hidden patterns and trends that might not be immediately apparent in raw data. Moreover, Power BI's integration with SQL supports real-time data analytics. This means that as data in the SQL database updates, the changes can be reflected in Power BI dashboards almost instantaneously. This feature is particularly beneficial for businesses that need to monitor data in real time to make timely and informed decisions.

Collaboration is another significant advantage of integrating SQL with Power BI. Power BI reports and dashboards can be shared across the organization, enabling stakeholders to access the insights derived from SQL data. Power BI supports role-based access, ensuring that data security is maintained while providing broad access to necessary insights. Furthermore, the integration of SQL and Power BI supports advanced analytics and machine learning. Power BI's AI features can be used to analyze SQL data, providing predictive analytics and identifying trends and anomalies. This capability enhances the strategic value of the data, enabling organizations to anticipate future trends and make proactive decisions.

The integration of SQL and Power BI creates a powerful data analysis and visualization environment. SQL's robust data management capabilities ensure that data

is well-structured and accurate, while Power BI's advanced visualization and analytical tools transform this data into actionable insights. This combination enhances the ability to make data-driven decisions, providing a significant strategic advantage to any organization. The seamless connectivity, real-time analytics, and collaborative features make the integration of SQL and Power BI a cornerstone of modern data analysis.

- **End-to-End Process:** Starting with data extraction and preparation in SQL, moving through visualization in Power BI, this integrated approach covers the entire spectrum of data handling and interpretation.
- **Comprehensive Insights:** The combination of detailed data querying and dynamic visualizations provides a holistic view of promotional effectiveness, uncovering both macro trends and micro-level insights.

4.5 Sampling Strategy

The study analyzed the entire dataset encompassing sales transactions and promotional activities during the specified campaign periods across all 50 supermarkets in Southern India. No sampling was conducted, ensuring a comprehensive analysis of all available data to provide accurate and reliable insights. By analyzing the complete dataset, the research aimed to capture the full range of variation and complexity within the data, enabling a thorough examination of promotional effectiveness and sales performance across different metrics.

CHAPTER 5

DATA ANALYSIS, FINDINGS AND INTERPRETATION

5.1 Data Preparation in SQL Workbench:

5.1.1 Create view for calculation of Promo_Price

```
1 • CREATE VIEW fact_events_pp AS
2   SELECT
3     fe.*,
4     CASE
5       WHEN fe.promo_type = 'BOGOF' THEN fe.base_price / 2 -- Buy One Get One Free, half the base price
6       WHEN fe.promo_type = '500 Cashback' THEN fe.base_price - 500 -- Cashback discount
7       WHEN fe.promo_type = '50% OFF' THEN fe.base_price * 0.5 -- 50% discount
8       WHEN fe.promo_type = '33% OFF' THEN fe.base_price * 0.67 -- 33% discount
9       WHEN fe.promo_type = '25% OFF' THEN fe.base_price * 0.75 -- 25% discount
10      ELSE fe.base_price -- Default to base price if no discount
11    END AS promo_price
12   FROM
13     fact_events fe;
```

Fig.5.1 Promo Price view

The creation of the `fact\_events\_pp` view in SQL Workbench constitutes a fundamental step in the data preparation process for AdHoc Insights within Atliq Mart's retail analytics framework. This view is crafted with a strategic purpose, aiming to refine the dataset and enhance its analytical capabilities by incorporating a derived column named `promo\_price`. This column serves as a key component in standardizing the representation of promotional prices across various promotional events, thereby facilitating more insightful analyses and robust decision-making processes.

The core objective behind introducing the `promo\_price` column lies in its ability to accurately capture the discounted prices of products during promotional periods. This becomes crucial for understanding the impact of promotions on sales performance, revenue generation, and customer behavior. Leveraging the SQL CASE statement, the

logic implemented within the view dynamically computes the promotional price based on the type of promotion associated with each event recorded in the `fact\_events` table.

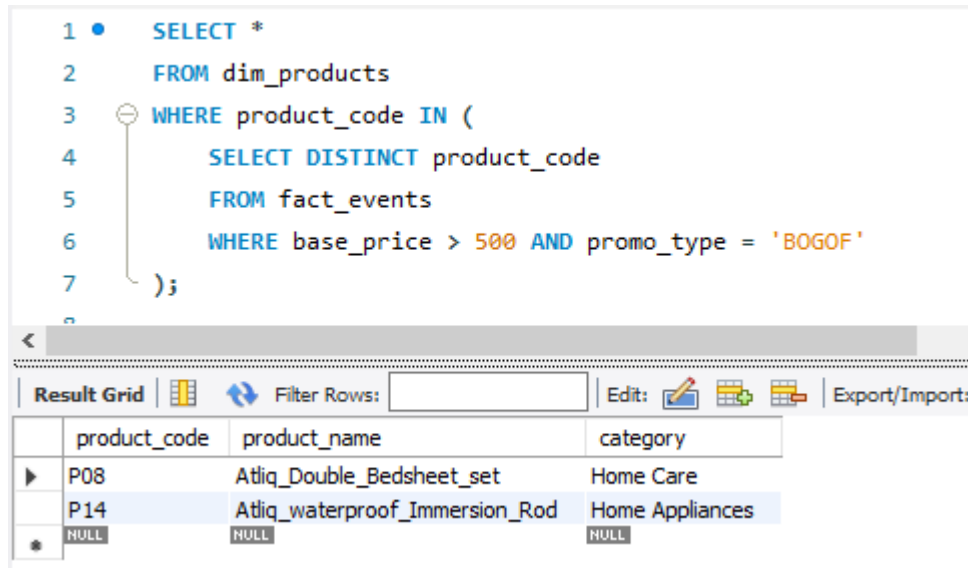
The conditional logic within the CASE statement meticulously evaluates the `promo\_type` column to determine the appropriate discounting mechanism for each promotional event. This approach ensures versatility and adaptability to diverse promotional strategies employed by Atliq Mart. For instance, when encountering a BOGOF (Buy One Get One Free) promotion, the `promo\_price` is set to half of the base price, aligning with the "Buy One Get One Free" offer. Similarly, for cashback promotions, a fixed amount is subtracted from the base price, while percentage-based discounts apply a specified percentage of the base price to compute the discounted price.

Moreover, the CASE statement gracefully handles scenarios where the `promo\_type` does not correspond to any predefined promotion types. In such cases, the logic gracefully defaults to the base price, ensuring the integrity and accuracy of the data representation. This comprehensive approach in data preparation guarantees that all products within the dataset are appropriately categorized, whether they are subject to promotional discounts or not.

By incorporating the `promo\_price` column into the `fact\_events\_pp` view, analysts gain access to a standardized and consistent representation of promotional prices across various promotional events. This standardization fosters a conducive environment for conducting in-depth analyses and deriving actionable insights regarding the effectiveness of different promotional strategies. Ultimately, the meticulous creation of the `fact\_events\_pp` view underscores Atliq Mart's commitment to harnessing the power of data-driven decision-making in optimizing promotional activities and driving sustained business growth.

5.2 AdHoc Insights

5.2.1 Products with a base price greater than 500 and promo type of 'BOGOF'



```
1 • SELECT *
2   FROM dim_products
3  WHERE product_code IN (
4      SELECT DISTINCT product_code
5      FROM fact_events
6      WHERE base_price > 500 AND promo_type = 'BOGOF'
7  );
```

product_code	product_name	category
P08	Atliq_Double_Bedsheet_set	Home Care
P14	Atliq_waterproof_Immersion_Rod	Home Appliances
NULL	NULL	NULL

Fig.5.2 Products with Base Price greater than 500

The query executed to retrieve the list of products with a base price greater than 500 and featured in the 'BOGOF' (Buy One Get One Free) promotional type from the dim_products table is a pivotal component of Atliq Mart's AdHoc Insights strategy. This query serves as a strategic tool for identifying high-value products that participate in promotional activities aimed at enhancing customer engagement and driving sales growth.

Upon execution, the query produces a concise yet insightful result set, showcasing the products that meet the specified criteria. The result set comprises two distinct product entries, each accompanied by essential details such as the product code, product name, and category. These details offer valuable insights into the nature and characteristics of the identified products, facilitating informed decision-making and strategic planning initiatives within Atliq Mart's retail ecosystem.

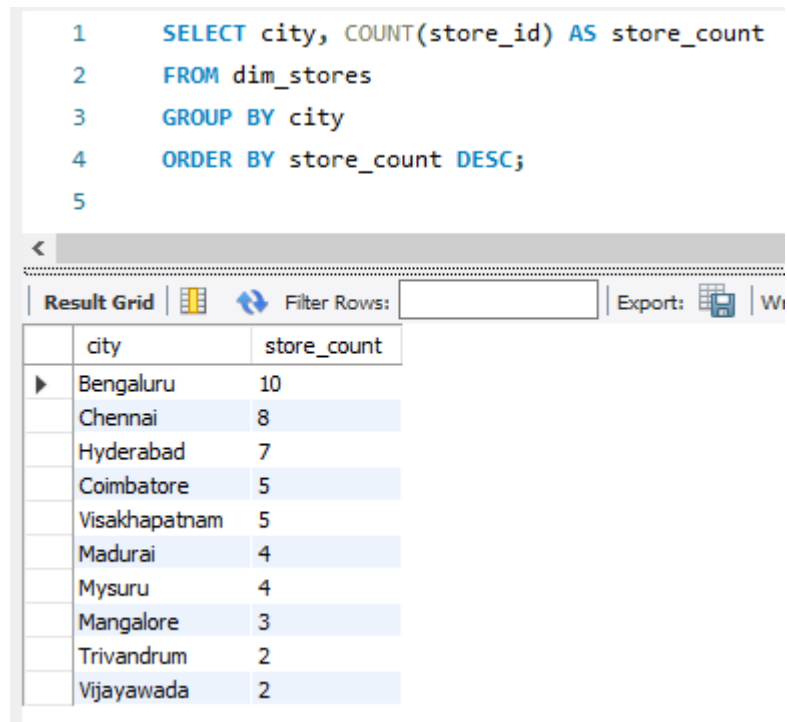
The first product listed, identified by the product code 'P08', corresponds to the "Atliq_Double_Bedsheet_set" within the 'Home Care' category. This product exemplifies Atliq Mart's offerings in the home care segment, catering to the diverse needs and preferences of its customer base. The inclusion of this product in the 'BOGOF' promotional type underscores Atliq Mart's strategic focus on incentivizing purchases and maximizing value proposition for its customers.

Similarly, the second product listed, denoted by the product code 'P14', and represents the "Atliq_waterproof_Immersion_Rod" belonging to the 'Home Appliances' category. This product exemplifies Atliq Mart's commitment to offering quality home appliances equipped with innovative features and functionalities. By featuring this product in the 'BOGOF' promotional type, Atliq Mart aims to stimulate demand and foster customer loyalty through compelling promotional offers and value-added benefits.

Overall, the query's execution and the resultant product list provide Atliq Mart with actionable insights into the intersection of product pricing and promotional strategies. By identifying products with a base price exceeding 500 and participating in 'BOGOF' promotions, Atliq Mart can strategically allocate resources, tailor marketing campaigns, and optimize inventory management to capitalize on emerging market trends and consumer preferences effectively.

Furthermore, the utilization of SQL subqueries within the query demonstrates Atliq Mart's proficiency in leveraging advanced analytical techniques to extract valuable insights from complex datasets. This approach exemplifies Atliq Mart's commitment to embracing data-driven decision-making processes and leveraging technology to drive sustainable business growth and competitive advantage in the dynamic retail landscape.

5.2.2 Store Distribution across Cities



The screenshot shows a SQL query editor with the following code:

```
1 SELECT city, COUNT(store_id) AS store_count
2 FROM dim_stores
3 GROUP BY city
4 ORDER BY store_count DESC;
5
```

Below the query editor is a 'Result Grid' showing the results of the query. The grid has two columns: 'city' and 'store_count'. The results are sorted in descending order of store count.

city	store_count
Bengaluru	10
Chennai	8
Hyderabad	7
Coimbatore	5
Visakhapatnam	5
Madurai	4
Mysuru	4
Mangalore	3
Trivandrum	2
Vijayawada	2

Fig.5.3 City wise store count

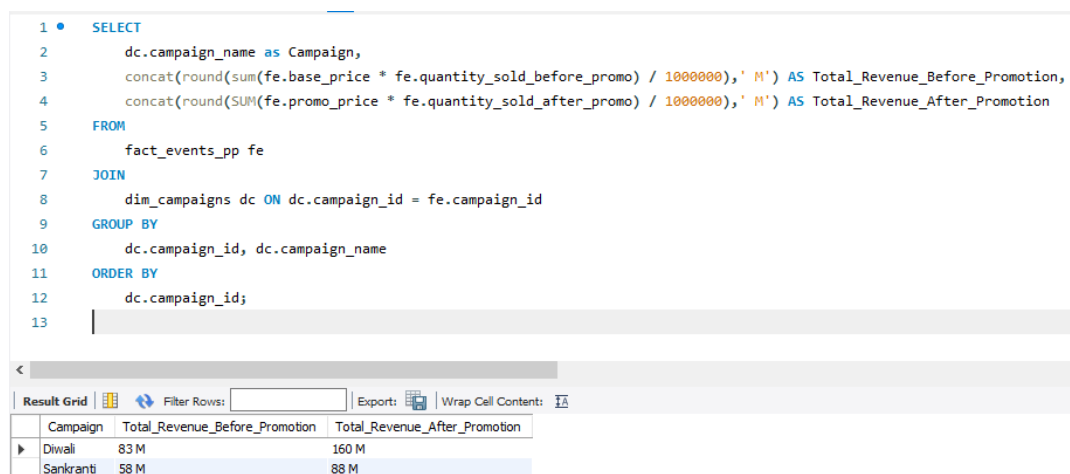
The query provided generates a report that offers an overview of the number of stores in each city, aiding in understanding the distribution of store presence across different cities. Let's break down the components of the query and its results:

The query selects two fields: "city" and the count of "store_id" for each city from the "dim_stores" table. It utilizes the GROUP BY clause to group the data by city, allowing for the calculation of the count of stores in each city. The COUNT () function is applied to the "store_id" column to determine the number of stores in each city. The result set is sorted in descending order of store counts using the ORDER BY clause, presenting the cities with the highest store presence at the top of the report.

Bengaluru has the highest number of stores, with a count of 10, making it the city with the most extensive store presence in the dataset. Chennai follows closely behind Bengaluru with 8 stores, indicating significant retail operations in this city as well.

Hyderabad ranks third with 7 stores, further highlighting its importance as a retail hub. Coimbatore and Visakhapatnam have equal store counts at 5 each, demonstrating a moderately high level of retail activity in these cities. Madurai and Mysuru each have 4 stores, while Mangalore has 3 stores, suggesting a relatively smaller but still notable presence of retail establishments. Trivandrum and Vijayawada have the fewest number of stores, with 2 each, indicating a relatively lower level of retail activity compared to other cities in the dataset. Overall, this report provides valuable insights into the distribution of stores across different cities, enabling Atliq Mart to better understand the geographical landscape of its retail operations and potentially identify opportunities for expansion or optimization in specific regions.

5.2.3 Promotional Campaign Impact Analysis: Revenue Comparison



```

1 • SELECT
2     dc.campaign_name as Campaign,
3     concat(round(sum(fe.base_price * fe.quantity_sold_before_promo) / 1000000), ' M') AS Total_Revenue_Before_Promotion,
4     concat(round(SUM(fe.promo_price * fe.quantity_sold_after_promo) / 1000000), ' M') AS Total_Revenue_After_Promotion
5 FROM
6     fact_events_pp fe
7 JOIN
8     dim_campaigns dc ON dc.campaign_id = fe.campaign_id
9 GROUP BY
10    dc.campaign_id, dc.campaign_name
11 ORDER BY
12    dc.campaign_id;
13

```

Campaign	Total_Revenue_Before_Promotion	Total_Revenue_After_Promotion
Diwali	83 M	160 M
Sankranti	58 M	88 M

Fig.5.4 Impact of Promotional Campaign

The query provided generates a comprehensive report that presents each campaign alongside the total revenue generated before and after the campaign, allowing for a detailed evaluation of the financial impact of promotional activities. Let's delve into the explanation and analysis of the query and its results:

Query Explanation: The query joins the "fact_events_pp" table with the "dim_campaigns" table based on the campaign_id to obtain relevant data for each campaign. It calculates the total revenue before the promotion by multiplying the base_price by the quantity_sold_before_promo and summing up these values. Similarly, it computes the total revenue after the promotion by multiplying the promo_price by the quantity_sold_after_promo and aggregating these values.

The ROUND () function is utilized to round the total revenue values to the nearest million for clarity, and the CONCAT () function is employed to append the unit "M" for millions to the rounded values. The GROUP BY clause groups the data by campaign_id and campaign_name to consolidate the revenue figures for each campaign. Finally, the ORDER BY clause arranges the results based on the campaign_id, ensuring a systematic presentation of the campaigns.

The report presents two campaigns: Diwali and Sankranti, along with their corresponding total revenue figures before and after the promotion, displayed in millions.

For the Diwali campaign, the total revenue before the promotion is 83 million, while the total revenue after the promotion significantly increases to 160 million, indicating a substantial boost in revenue attributed to the promotional activities during the Diwali period.

Similarly, for the Sankranti campaign, the total revenue before the promotion stands at 58 million, and post-promotion, it rises to 88 million, demonstrating a notable enhancement in revenue as a result of the promotional efforts undertaken during the Sankranti festival. Overall, the report provides valuable insights into the effectiveness of promotional campaigns in driving revenue growth, enabling stakeholders to assess the

financial impact of these initiatives and make informed decisions regarding future promotional strategies and resource allocations.

5.2.4 Diwali Campaign Impact Analysis: Category-wise Incremental Sold Quantity (ISU%) and Rankings



```

1 WITH ISU_Percent AS (
2     SELECT dp.category, dc.campaign_name,
3         ROUND((SUM(fe.quantity_sold_after_promo) - SUM(fe.quantity_sold_before_promo)) / NULLIF(SUM(fe.quantity_sold_before_promo), 0)) * 100, 1)
4     AS isu_percentage
5     FROM dim_products dp
6     JOIN fact_events fe ON dp.product_code = fe.product_code
7     JOIN dim_campaigns dc ON fe.campaign_id = dc.campaign_id
8     GROUP BY dp.category, dc.campaign_name
9 )
10 SELECT diwali.category, CONCAT(diwali.isu_percentage, ' %') AS "Diwali ISU%", CONCAT(sankranti.isu_percentage, ' %') AS "Sankranti ISU%"
11 FROM (SELECT category, isu_percentage FROM ISU_Percent WHERE campaign_name = 'Diwali') diwali
12 JOIN (SELECT category, isu_percentage FROM ISU_Percent WHERE campaign_name = 'Sankranti') sankranti
13 ON diwali.category = sankranti.category
14 ORDER BY sankranti.isu_percentage DESC;
15

```

category	Diwali ISU%	Sankranti ISU%
Home Appliances	244.2 %	275.5 %
Home Care	79.6 %	162.4 %
Grocery & Staples	18.0 %	158.2 %
Combo1	202.4 %	125.3 %
Personal Care	31.1 %	11.5 %

Fig.5.5 Impact of Diwali Campaign

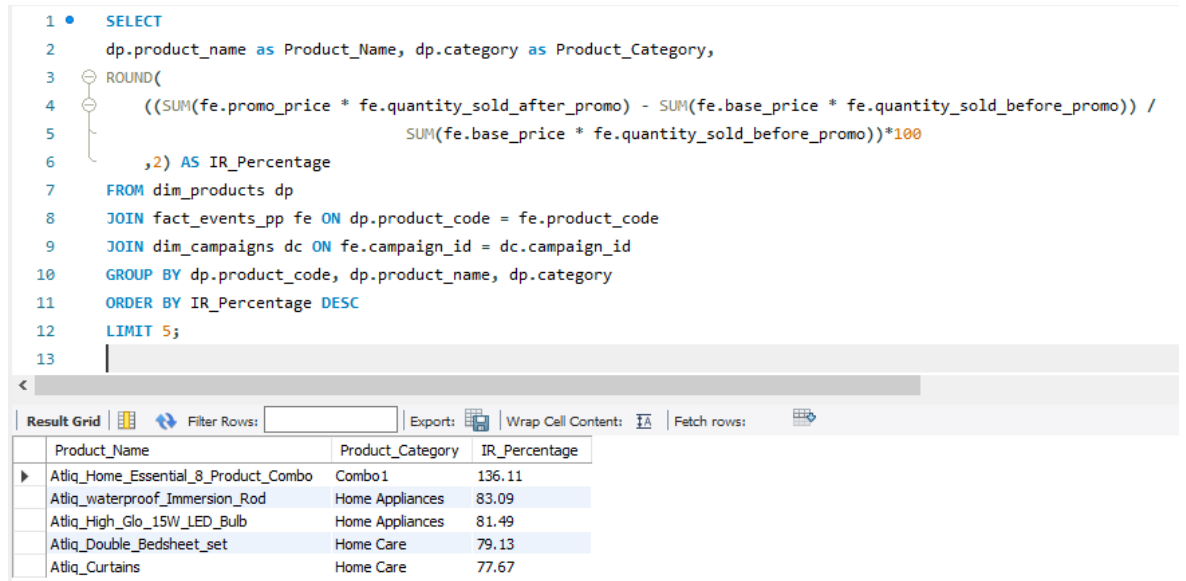
The provided query generates a comprehensive report that calculates the Incremental Sold Quantity Percentage (ISU%) for each product category during the Diwali campaign. Additionally, it ranks the categories based on their ISU%, allowing for a detailed assessment of the campaign's impact on incremental sales across different categories. The query leverages a window function, RANK(), to assign a rank order to each category based on their ISU%. The ISU% is calculated as the percentage increase or decrease in quantity sold (after promo) compared to quantity sold (before promo). It joins the "dim_products" table with the "fact_events" table based on the product_code and further connects with the "dim_campaigns" table based on the campaign_id to retrieve relevant data for the Diwali campaign. The WHERE clause filters the data to include only the records related to the Diwali campaign. The query then groups the data by category using

the GROUP BY clause to calculate the ISU% for each category. Finally, the results are ordered based on the rank_order derived from the RANK() function, providing a ranked list of categories based on their ISU%.

The report presents the categories ranked by their ISU%, showcasing the percentage increase or decrease in quantity sold during the Diwali campaign compared to the pre-promotion period. Home Appliances emerge as the category with the highest ISU%, indicating a significant surge in sales during the Diwali campaign, with an ISU% of 244.2%. This suggests that promotional activities targeted towards home appliances were particularly effective in driving incremental sales. Combo1 follows closely with an ISU% of 202.4%, signifying a substantial increase in sales for bundled products during the campaign period. Home Care and Personal Care categories also experienced notable ISU% of 79.6% and 31.1%, respectively, highlighting a positive impact on sales in these categories due to the Diwali promotions.

Grocery & Staples exhibit a comparatively lower ISU% of 18.0%, indicating a relatively modest increase in sales compared to other categories, which may require further analysis to optimize promotional strategies for this category. Overall, the report provides valuable insights into the effectiveness of the Diwali campaign across different product categories, enabling stakeholders to tailor future promotional efforts and allocate resources effectively to maximize incremental sales potential across the product portfolio.

5.2.5 Top 5 Products by Incremental Revenue Percentage (IR%) Across Campaigns



```

1 • SELECT
2   dp.product_name as Product_Name, dp.category as Product_Category,
3   ROUND(
4     ((SUM(fe.promo_price * fe.quantity_sold_after_promo) - SUM(fe.base_price * fe.quantity_sold_before_promo)) /
5      SUM(fe.base_price * fe.quantity_sold_before_promo))*100
6     ,2) AS IR_Percentage
7 FROM dim_products dp
8 JOIN fact_events_pp fe ON dp.product_code = fe.product_code
9 JOIN dim_campaigns dc ON fe.campaign_id = dc.campaign_id
10 GROUP BY dp.product_code, dp.product_name, dp.category
11 ORDER BY IR_Percentage DESC
12 LIMIT 5;
13

```

Product_Name	Product_Category	IR_Percentage
Atliq_Home_Essential_8_Product_Combo	Combo1	136.11
Atliq_waterproof_Immersion_Rod	Home Appliances	83.09
Atliq_High_Glo_15W_LED_Bulb	Home Appliances	81.49
Atliq_Double_Bedsheet_set	Home Care	79.13
Atliq_Curtains	Home Care	77.67

Fig.5.6 Top 5 Products by Incremental Revenue Percentage

The provided query generates a report featuring the Top 5 products ranked by Incremental Revenue Percentage (IR%) across all campaigns. This analysis is crucial for identifying the most successful products in terms of generating incremental revenue throughout various promotional campaigns, thereby aiding in product optimization and strategic decision-making. Let's delve into the explanation and analysis of the query and its results:

The query utilizes joins between the "dim_products," "fact_events_pp," and "dim_campaigns" tables based on their respective keys (product_code, campaign_id) to retrieve relevant data for the analysis. It calculates the IR% for each product by subtracting the total revenue generated before the promotion from the total revenue generated after the promotion, divided by the total revenue before the promotion, and then multiplied by 100 to get the percentage increase. The results are grouped by product_code, product_name, and category using the GROUP BY clause to ensure

distinct products are included in the analysis. Finally, the results are ordered in descending order based on IR% and limited to the top 5 products using the LIMIT clause.

The report presents the top 5 products ranked by their IR%, providing insights into the products that have contributed the most to incremental revenue across all campaigns. The Atliq Home Essential 8 Product Combo emerges as the top-performing product with an IR% of 136.11%, indicating a significant increase in revenue generated after promotions compared to before. Home Appliances feature prominently in the top-ranked products, with both the Atliq Waterproof Immersion Rod and the Atliq High Glo 15W LED Bulb securing positions in the top 5, showcasing their effectiveness in driving incremental revenue. Home Care products such as the Atliq Double Bedsheet Set and Atliq Curtains also demonstrate strong performance, with IR% values of 79.13% and 77.67%, respectively.

The analysis highlights the importance of these top-performing products in driving incremental revenue across campaigns, providing valuable insights for optimizing promotional strategies and focusing resources on products with the highest revenue potential. Overall, the report aids stakeholders in identifying key products that contribute significantly to incremental revenue, enabling informed decision-making and targeted promotional efforts to maximize overall revenue growth.

5.3 Integration of Power BI with SQL

Integration of Power BI and SQL This integration combines the powerful visualization capabilities of Power BI with the data retrieval and manipulation features of SQL Workbench, enabling users to extract valuable insights from SQL databases and present them in a visually compelling manner. By integrating these two platforms, organizations

can streamline their data analytics workflows, improve decision-making processes, and gain a deeper understanding of their business operations.

Power BI, developed by Microsoft, is a leading business intelligence tool known for its user-friendly interface, advanced visualization options, and seamless integration with various data sources. It empowers users to create interactive dashboards, reports, and data visualizations that facilitate data-driven decision-making across the organization. On the other hand, SQL Workbench provides a robust environment for database management, query execution, and data manipulation, making it a preferred choice for SQL developers and analysts.

The integration between Power BI and SQL Workbench bridges the gap between data storage and data visualization, allowing users to access SQL data directly within Power BI and transform it into actionable insights. This integration eliminates the need for manual data extraction and manipulation, reducing the time and effort required to generate meaningful reports and analyses. Additionally, it enables users to leverage the full potential of SQL Workbench's database querying capabilities while harnessing Power BI's intuitive visualization tools to communicate key findings effectively.

By providing a seamless connection between data storage and visualization, this integration empowers users to extract actionable insights from their data and gain a competitive edge in today's data-driven business landscape.

Setting up the integration between Power BI and SQL Workbench is a crucial step in enabling data analysis and visualization within Power BI using data stored in SQL databases. To begin, users need to ensure that they have the necessary connectors or drivers installed in Power BI to establish a connection with SQL Workbench.

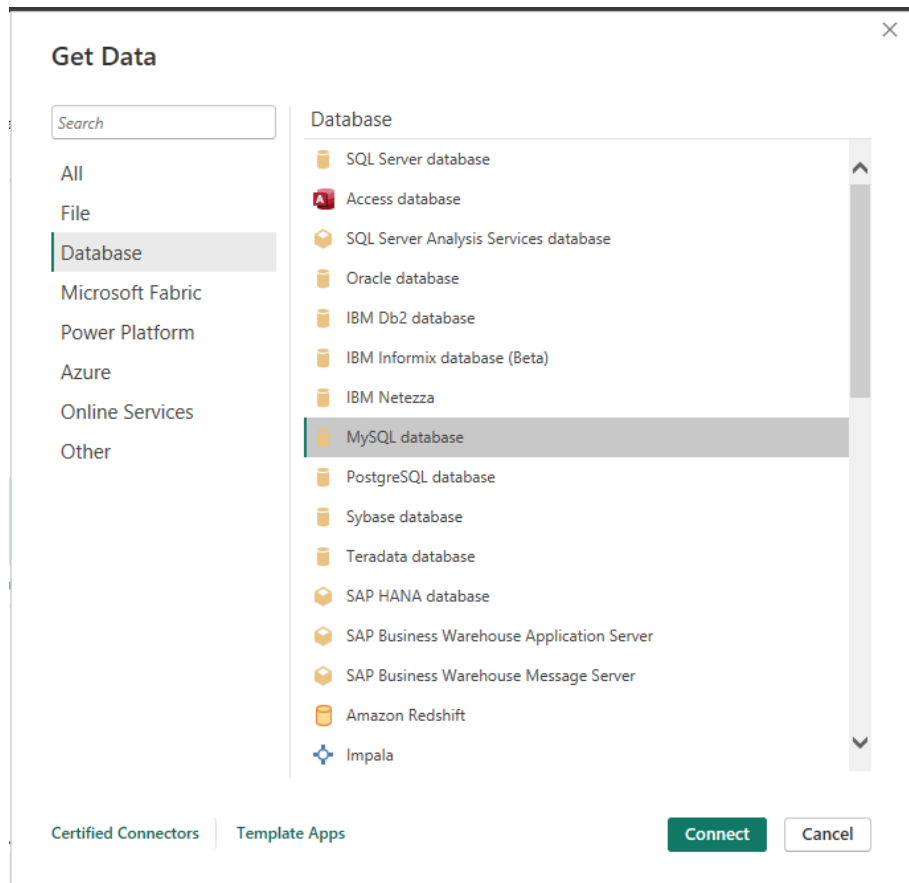


Fig.5.7 MySQL and Power BI Integration

This typically involves navigating to the "Get Data" option in Power BI Desktop and selecting the "Database" category, followed by choosing "SQL Server" as the database type as shown below.

If the SQL Workbench instance is based on SQL Server, this option is appropriate. Users may be prompted to install ODBC drivers for SQL Server connectivity, which facilitate communication between Power BI and the SQL Server database. Once the necessary drivers are installed, users can proceed to establish a connection to the SQL database within Power BI. This involves providing connection details such as the server name or IP address of the SQL Server where SQL Workbench is hosted, as well as specifying the authentication method.

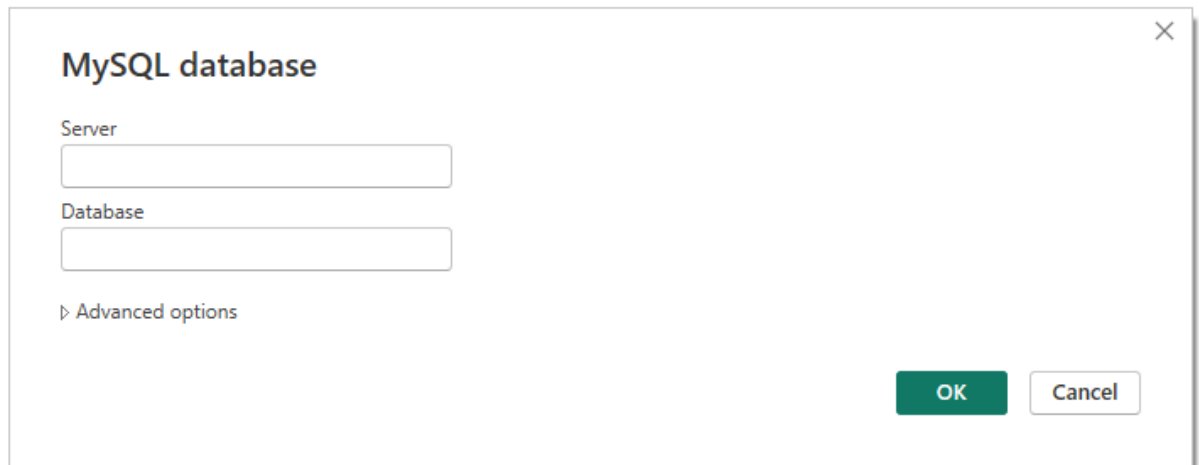


Fig.5.8 MySQL Database Login

Depending on the SQL Server configuration, users can choose between Windows Authentication and SQL Server Authentication. If SQL Server Authentication is selected, users must enter the username and password associated with their SQL Workbench credentials. Additionally, users may specify a specific database or schema to connect to within the SQL Server if multiple databases or schemas are present. Testing the connection is essential to ensure that Power BI can successfully connect to the SQL Server database using the provided credentials.

Importing data into Power BI from the connected SQL database is the next step. After establishing a successful connection, Power BI will display a list of tables and views available in the SQL Server database. Users can select the desired tables or views containing the data they wish to analyse in Power BI as shown below.

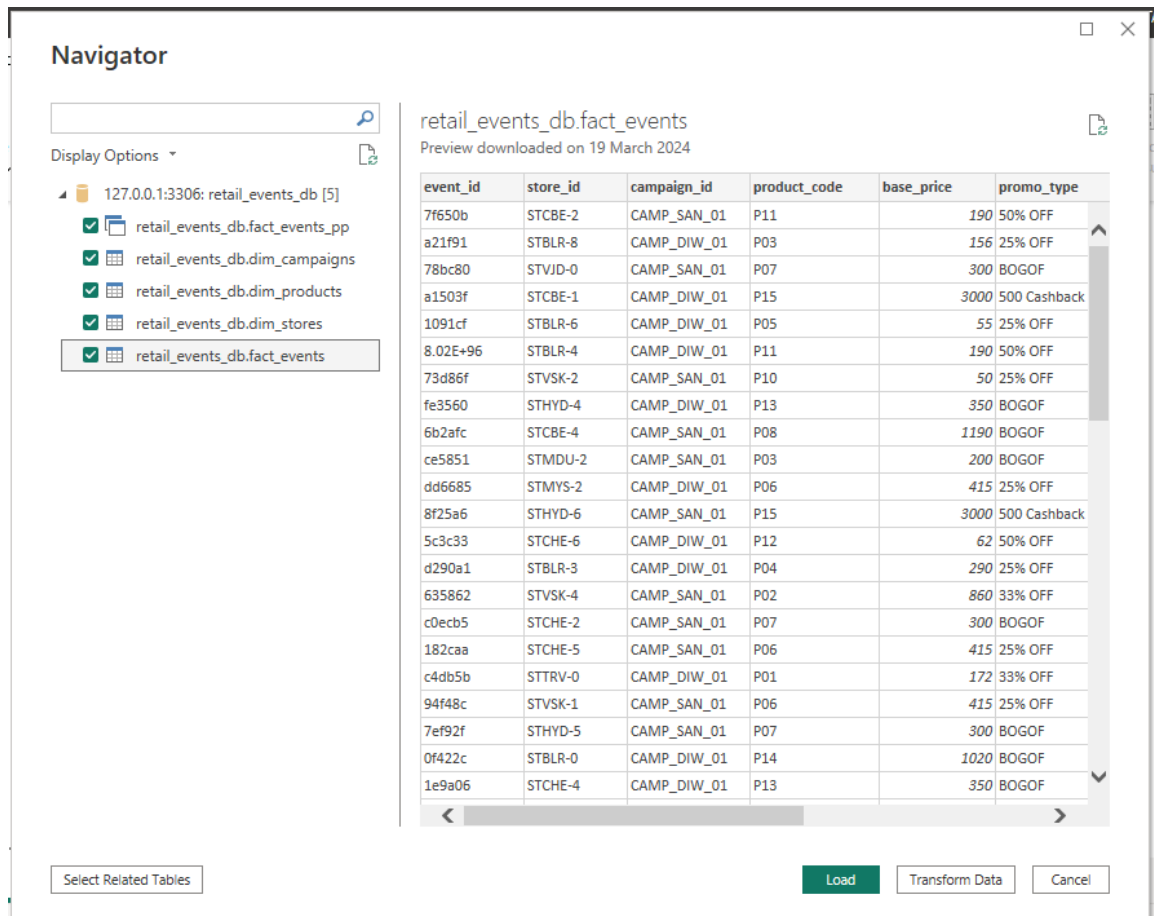


Fig.5.9 Data Load Preview in Power BI

Previewing the data before loading ensures that it meets the requirements. Clicking the "Load" button imports the selected data into Power BI. Alternatively, users can choose the "Transform Data" option to perform data cleansing and transformation operations before loading, ensuring that the imported data is ready for analysis and visualization in Power BI. After loading the data, the next step is to create the Key Performance Indicators (KPIs) that are required for further analysis.

5.4 Key Performance Indicators

5.4.1 Revenue Before Promotion

Revenue Before Promotion is a key performance indicator that measures the total revenue generated from product sales before the implementation of any promotional campaign or discount. It provides insights into the baseline revenue generated by the company's products without any promotional activities influencing consumer purchasing behaviour. The provided DAX formula calculates Revenue Before Promotion by iterating over each row in the 'fact_events_pp' table and multiplying the base price of each product by the quantity sold before the promotion. It then sums up these individual revenue values to obtain the total revenue generated from product sales before any promotional discounts or campaigns are applied.

Including Revenue Before Promotion as a key performance indicator in your project report allows stakeholders to understand the baseline revenue generated by the company's products before any promotional efforts. It serves as a reference point for evaluating the impact of promotional campaigns on overall revenue generation and assessing the effectiveness of marketing strategies in driving sales.

DAX Formula:

Revenue Before Promotion =

```
SUMX(  
    'fact_events_pp',  
    'fact_events_pp'[base_price] * 'fact_events_pp'[quantity_sold_before_promo]  
)
```

5.4.2 Revenue After Promotion

Revenue After Promotion represents the total revenue generated from product sales after the implementation of promotional campaigns. It is a crucial metric for evaluating the financial performance and effectiveness of promotional activities in driving revenue growth.

The provided DAX formula calculates Revenue After Promotion by iterating through each row of the 'fact_events_pp' table and multiplying the promotional price per unit ('promo_price') by the quantity of products sold after the promotion ('quantity_sold_after_promo'). It then sums up these calculated values across all rows to obtain the total revenue generated after the promotional campaigns.

Including Revenue After Promotion as a key performance indicator in your project report provides insights into the financial impact of promotional activities on overall revenue generation. It allows stakeholders to assess the effectiveness of promotions in driving sales and revenue growth, enabling them to make informed decisions regarding marketing strategies and investment in future promotional campaigns.

DAX Formula:

Revenue After Promotion =

```
SUMX(  
    'fact_events_pp',  
    'fact_events_pp'[promo_price] * 'fact_events_pp'[quantity_sold_after_promo]  
)
```

5.4.3 Incremental Sold Units

The Incremental Sold Units (ISU) is a key performance indicator that measures the change in the quantity of products sold after a promotional campaign compared to before the campaign. It helps in evaluating the effectiveness of promotions in driving additional

sales. The DAX formula provided calculates the ISU by subtracting the total quantity of products sold before the promotional campaign from the total quantity sold after the campaign. This difference represents the incremental or additional units sold directly attributed to the promotion. Including this KPI in your project report provides insights into the impact of promotional activities on sales performance, allowing stakeholders to assess the effectiveness of marketing strategies and make informed decisions regarding future promotional efforts.

DAX Formula:

Incremental Sold Units = `SUM(fact_events_pp[quantity_sold_after_promo]) - SUM(fact_events_pp[quantity_sold_before_promo])`

5.4.4 Incremental Revenue:

Incremental Revenue is a crucial metric used to assess the financial impact of promotional campaigns on sales revenue. It quantifies the difference in revenue generated after a promotional campaign compared to revenue generated before the campaign.

The provided DAX formula calculates Incremental Revenue by subtracting the total revenue generated before the promotional campaign from the total revenue generated after the campaign. This calculation captures the additional revenue attributed to the promotion.

Including Incremental Revenue as a key performance indicator in your project report allows stakeholders to understand the financial effectiveness of promotional activities. It provides insights into the direct impact of promotions on revenue generation, facilitating data-driven decision-making and strategy optimization.

DAX Formula:

Incremental Revenue = `[Revenue After Promotion] - [Revenue Before Promotion]`

5.4.5 ISU Percent

ISU Percent (Incremental Sold Units Percentage) is a metric used to measure the percentage increase or decrease in the quantity of products sold after a promotional campaign compared to before the campaign. It helps to evaluate the effectiveness of promotions in driving sales volume.

The provided DAX formula calculates ISU Percentage by first subtracting the total quantity of products sold before the promotion from the total quantity sold after the promotion. This difference represents the incremental sold units. Then, it divides this value by the total quantity sold before the promotion to obtain the relative change in sales volume. Finally, the result is multiplied by 100 and rounded to two decimal places to express it as a percentage. Including ISU Percentage as a key performance indicator in your project report provides insights into the effectiveness of promotional campaigns in driving sales volume. It allows stakeholders to assess the impact of promotions on consumer behaviour and product demand, enabling informed decision-making in marketing and sales strategies.

DAX Formula:

```
ISU_Percentagee = ROUND(((SUM(fact_events_pp[quantity_sold_after_promo])-  
SUM(fact_events_pp[quantity_sold_before_promo]))/SUM(fact_events_pp[quantity_sold_bef  
ore_promo]))*100,2)
```

5.4.6 IR Percent

IR Percent (Incremental Revenue Percentage) is a key performance indicator used to measure the percentage increase or decrease in revenue generated after a promotional campaign compared to before the campaign. It provides valuable insights into the financial impact of promotions on overall revenue.

The provided DAX formula calculates IR Percent by first subtracting the total revenue generated before the promotion from the total revenue generated after the promotion. This difference represents the incremental revenue. Then, it divides this incremental revenue by the total revenue before the promotion to obtain the relative change in revenue. Finally, the result is multiplied by 100 to express it as a percentage.

Including IR Percent as a key performance indicator in your project report enables stakeholders to assess the effectiveness of promotional campaigns in driving revenue growth. It helps in evaluating the return on investment (ROI) of promotions and identifying which campaigns contribute the most to revenue enhancement. This information is essential for optimizing promotional strategies and maximizing profitability.

DAX Formula:

IR Percent =

```
DIVIDE(  
    ([Revenue After Promotion]-[Revenue Before Promotion]),[Revenue Before Promotion]  
) * 100  
  
DIVIDE(__Numerator, __Denominator)  
)
```

5.4.7 Store Performance Index

The Store Performance Index is a metric used to rank stores based on their performance in terms of generating revenue from product sales after promotional activities. It helps evaluate and compare the effectiveness of different stores in driving sales and revenue growth through promotional campaigns.

The provided DAX formula calculates the Store Performance Index by first iterating through each store ID ('store_id') in the 'dim_stores' table. It then calculates the total revenue generated by each store after promotions by multiplying the quantity of products sold after the promotion ('quantity_sold_after_promo') by the promotional price per unit ('promo_price') for each transaction in the 'fact_events_pp' table. The calculated revenue values are summed up for each store.

Once the total revenue values are obtained for each store, the RANKX function is used to assign a rank to each store based on its revenue performance. The ALL function is employed to ensure that all store IDs are considered for ranking, regardless of any filters applied to the data. The stores are ranked in descending order, with the store generating the highest revenue from promotions receiving the highest rank. Including the Store Performance Index in your project report provides insights into the relative performance of different stores in driving revenue through promotional activities. It helps identify top-performing stores that contribute significantly to overall sales and revenue, as well as areas for improvement in stores that may have lower performance rankings. This information enables stakeholders to optimize resource allocation, target specific stores for performance improvement initiatives, and make data-driven decisions to enhance overall store performance.

DAX Formula:

StorePerformanceIndex =

```
RANKX(  
    ALL(dim_stores[store_id]),  
    CALCULATE(  
        SUMX(  
            RELATEDTABLE('fact_events_pp'),  
            [quantity_sold_after_promo] * [promo_price] ) ) )
```

5.4.8 City Performance Index

The City Performance Index is a metric used to rank cities based on their performance in terms of generating revenue from product sales after promotional activities. It provides insights into which cities are the most successful in driving revenue growth through promotions, allowing businesses to identify key markets and tailor their marketing strategies accordingly.

The provided DAX formula calculates the City Performance Index by first iterating through each city in the 'dim_stores' table using the ALL function. Then, for each city, it calculates the total revenue generated from product sales after promotions by summing up the product of the quantity sold after the promotion and the promotional price per unit for each transaction. This calculation is performed using the SUMX and RELATEDTABLE functions. Once the total revenue is calculated for each city, the RANKX function assigns a rank to each city based on its revenue performance. Cities with higher total revenue will receive lower rank numbers, indicating better performance, while cities with lower revenue will receive higher rank numbers.

Including the City Performance Index in your project report provides valuable insights into the effectiveness of promotional activities across different cities. It helps businesses identify high-performing markets, assess the impact of promotions on revenue generation, and make data-driven decisions to optimize their marketing strategies and allocate resources effectively.

DAX Formula:

CityPerformanceIndex =

```
RANKX(  
    ALL(dim_stores[city]),  
    CALCULATE(  

```

```
SUMX(
    RELATEDTABLE('fact_events_pp'),
    [quantity_sold_after_promo] * [promo_price]
) ) )
```

5.4.9 Average Order Value Before Promo

The Average Order Value Before Promo is a key performance indicator (KPI) used to measure the average value of orders placed by customers before any promotional discounts or offers are applied. It provides insights into the typical spending behaviour of customers prior to engaging with promotional campaigns.

The provided DAX formula calculates the Average Order Value Before Promo by first multiplying the base price of each product by the quantity sold before the promotion for each transaction using the SUMX function. This calculates the total revenue generated from product sales before any promotions are applied. Then, it divides the total revenue by the total number of transactions (count of rows) in the 'fact_events_pp' table using the COUNTROWS function to compute the average order value.

Including the Average Order Value Before Promo in your project report allows stakeholders to understand the baseline spending patterns of customers before they are influenced by promotional activities. It helps businesses assess the impact of promotions on customer spending behaviour and evaluate the effectiveness of promotional campaigns in driving higher order values. Additionally, it provides insights into customer preferences and purchasing habits, enabling businesses to tailor their promotional strategies more effectively to meet customer needs and maximize revenue generation.

DAX Formula:

AverageOrderValueBeforePromo =
 DIVIDE(

```
SUMX(fact_events_pp, [base_price] * [quantity_sold_before_promo]),  
COUNTROWS(fact_events_pp)  
)
```

5.4.10 Average Order Value After Promo

The Average Order Value After Promotion is a key performance indicator that measures the average value of orders placed by customers after promotional activities. It provides insights into the effectiveness of promotions in driving higher purchase amounts per transaction.

The provided DAX formula calculates the Average Order Value After Promotion by first iterating through each row in the 'fact_events_pp' table using the SUMX function. For each row, it multiplies the promotional price per unit by the quantity sold after the promotion to calculate the total revenue generated from that transaction. Then, it sums up the total revenue from all transactions.

Next, the total revenue is divided by the total number of rows in the 'fact_events_pp' table using the COUNTROWS function. This division yields the average order value after the promotion, representing the average amount spent by customers per transaction after taking into account promotional discounts or incentives.

Including the Average Order Value After Promotion in your project report provides valuable insights into customer purchasing behaviour and the effectiveness of promotional strategies in driving higher transaction values. It helps businesses evaluate the impact of promotions on revenue generation and make informed decisions to optimize their marketing efforts and enhance overall sales performance.

DAX Formula:

AverageOrderValueAfterPromo =

```
DIVIDE(  
    SUMX(fact_events_pp, [promo_price] * [quantity_sold_after_promo]),  
    COUNTROWS(fact_events_pp)  
)
```

5.4.11 Incremental Average Order Value

The Incremental Average Order Value is a crucial metric that evaluates the change in the average value of orders placed by customers before and after promotional activities. It helps businesses understand the impact of promotions on increasing or decreasing the average transaction amount per customer.

The provided formula calculates the Incremental Average Order Value by subtracting the Average Order Value Before Promotion from the Average Order Value After Promotion. The Average Order Value Before Promotion represents the average amount spent by customers per transaction before any promotional discounts or incentives are applied. On the other hand, the Average Order Value After Promotion reflects the average transaction amount after accounting for promotional discounts or incentives. By subtracting the Average Order Value Before Promotion from the Average Order Value After Promotion, the Incremental Average Order Value quantifies the change in the average transaction amount attributed to promotional activities. A positive value indicates an increase in the average transaction amount post-promotion, while a negative value suggests a decrease.

Including the Incremental Average Order Value in your project report provides valuable insights into the effectiveness of promotions in driving changes in customer spending behaviour. It helps businesses assess the success of promotional strategies in

encouraging customers to spend more per transaction, ultimately leading to increased revenue and improved profitability.

DAX Formula:

Incremental Average Order Value = $[AverageOrderValueAfterPromo] - [AverageOrderValueBeforePromo]$

5.5 Store Performance Analysis

5.5.1 Top 10 Stores by Incremental Revenue (IR) from Promotions

store_id	city	Incremental Revenue
STMYS-1	Mysuru	3.63 Million
STCHE-4	Chennai	3.54 Million
STBLR-0	Bengaluru	3.51 Million
STBLR-7	Bengaluru	3.40 Million
STMYS-3	Mysuru	3.36 Million
STCHE-7	Chennai	3.34 Million
STBLR-6	Bengaluru	3.29 Million
STCHE-3	Chennai	3.16 Million
STMDU-0	Madurai	2.83 Million
STBLR-3	Bengaluru	2.78 Million
Total		32.83 Million

Table.5.1 Top 10 Stores by Incremental Revenue

The analysis of the top 10 stores in terms of Incremental Revenue (IR) sheds light on several key insights regarding the effectiveness of promotions and the performance of store locations within AtliQ Mart's retail network.

Firstly, it's notable that the top-performing stores are distributed across multiple cities, with Mysuru, Chennai, and Bengaluru featuring prominently. This suggests that the success of promotional campaigns is not confined to a single geographical region but rather extends across different markets where AtliQ Mart operates.

Secondly, the variations in Incremental Revenue among the top-performing stores underscore the significance of local market dynamics and consumer behaviour. Stores in certain cities may exhibit higher Incremental Revenue due to factors such as population density, consumer preferences, or the competitive landscape within the retail sector. Additionally, the analysis highlights the importance of store-level execution and management in driving promotional effectiveness. Stores with strong sales performance during promotional periods likely have effective sales teams, merchandising strategies, and customer engagement tactics that contribute to their success.

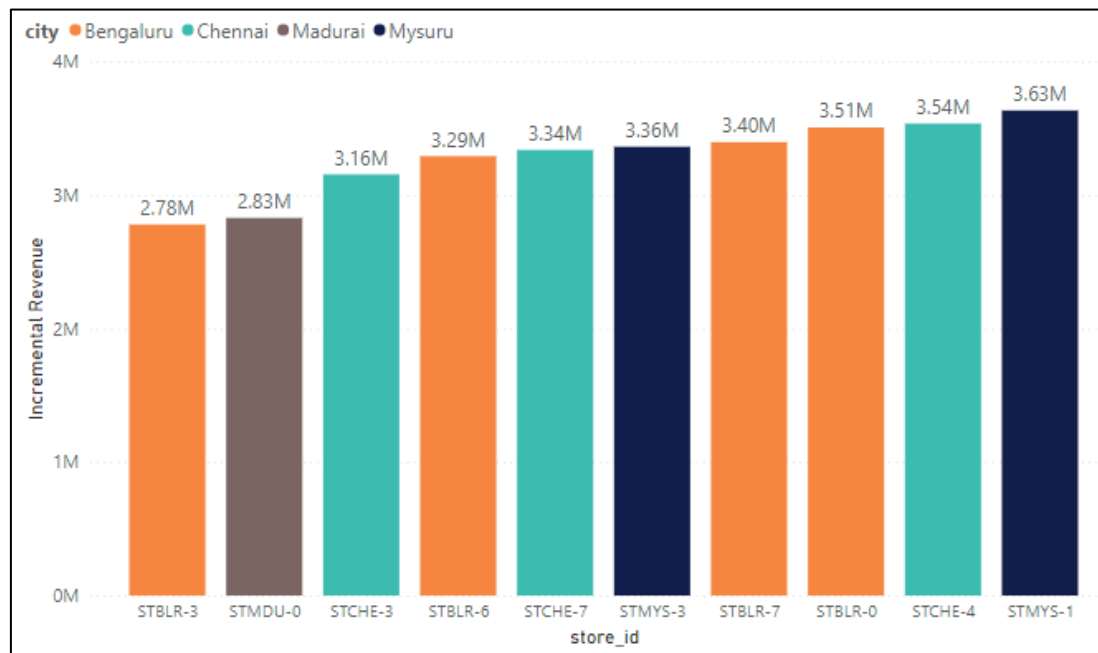


Fig.5.10 Top 10 Stores by Incremental Revenue

Moreover, the cumulative Incremental Revenue from the top 10 stores underscores the substantial impact that promotions have on overall sales performance. By identifying and leveraging the strengths of these top-performing stores, AtliQ Mart can refine its promotional strategies and replicate success across other store locations to maximize revenue generation. Furthermore, understanding the factors that contribute to the success of promotions in specific store locations can inform targeted marketing efforts and

resource allocation. For instance, insights gained from analysing the top-performing stores can guide decisions related to product assortment, pricing strategies, and promotional tactics tailored to the preferences and behaviours of local consumers.

In conclusion, the analysis of Incremental Revenue across the top 10 stores provides valuable insights into the effectiveness of promotions and the performance of store locations within AtliQ Mart's retail network. By leveraging these insights, AtliQ Mart can optimize its promotional strategies, enhance store-level performance, and drive sustainable growth in sales and revenue across its retail operations.

5.5.2 Bottom 10 Stores by Incremental Sold Units (ISU) During Promotions

Store_id	City	Incremental Sold Units
STVJD-0	Vijayawada	3046
STCBE-4	Coimbatore	2927
STMLR-1	Mangalore	2784
STVJD-1	Vijayawada	2763
STTRV-0	Trivandrum	2733
STMLR-2	Mangalore	2664
STTRV-1	Trivandrum	2604
STVSK-4	Visakhapatnam	2469
STVSK-3	Visakhapatnam	2209
STMLR-0	Mangalore	1952
Total		26151

Table.5.2 Bottom 10 Stores by Incremental Sold Units

Analyzing the bottom 10 stores in terms of Incremental Sold Units (ISU) during the promotional period reveals several noteworthy observations. Vijayawada (STVJD-0) despite being a prominent city, the store in Vijayawada exhibits lower-than-average incremental sold units, suggesting potential challenges in engaging customers during

promotional campaigns. Factors such as competition from local retailers or insufficient marketing efforts may contribute to this trend. Coimbatore (STCBE-4) known for its bustling retail landscape, surprisingly records below-average incremental sold units. This could indicate a mismatch between promotional strategies and customer preferences in this market. Analyzing consumer behaviour and preferences specific to Coimbatore could provide insights into improving promotional effectiveness.

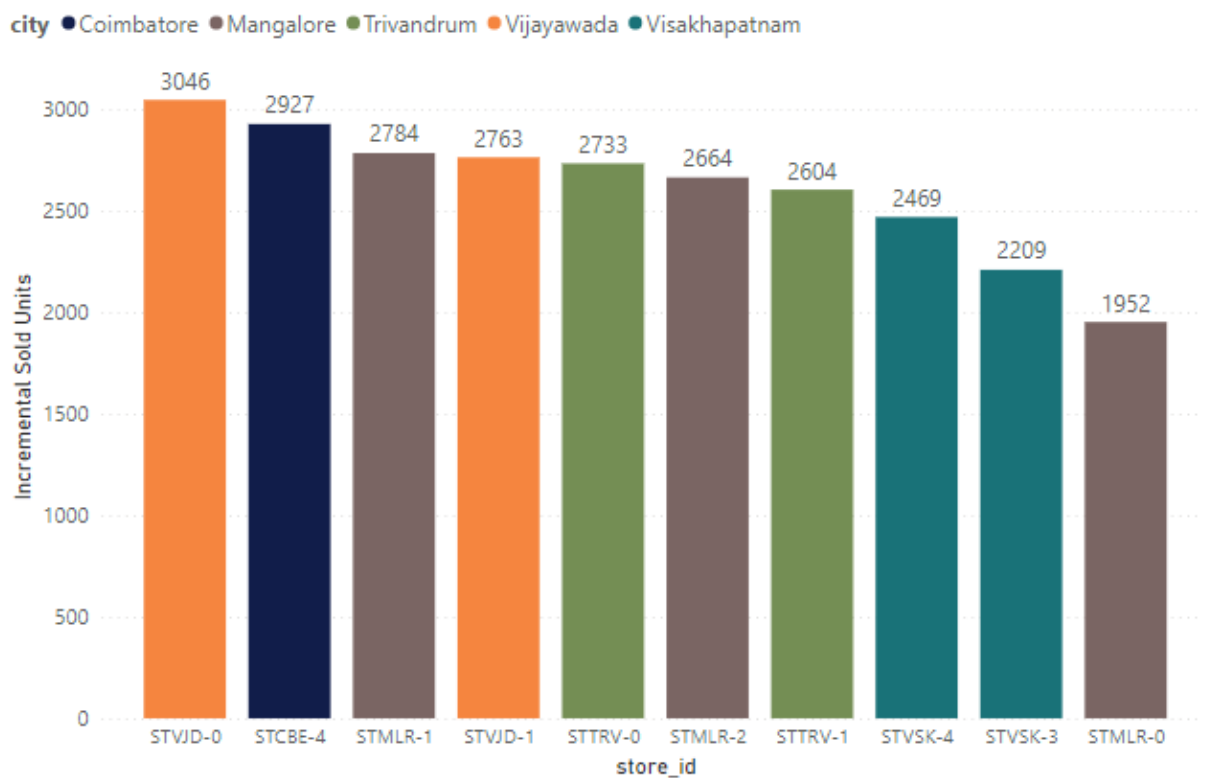


Fig.5.11 Bottom 10 Stores by Incremental Sold Units

Mangalore (STMLR-1 and STMLR-2) despite being a significant urban center, the stores in Mangalore show lower-than-expected incremental sold units. Possible factors contributing to this trend include stiff competition from local and regional retailers, as well as potential challenges in effectively communicating promotional offers to customers. Trivandrum (STTRV-0 and STTRV-1) stores in Trivandrum demonstrate below-average incremental sold units, indicating a need for tailored promotional

strategies to resonate with the local customer base. Understanding the unique consumer preferences and market dynamics of Trivandrum could help in designing more effective promotional campaigns.

Visakhapatnam (STVSK-4 and STVSK-3) despite being a bustling coastal city, stores in Visakhapatnam show lower incremental sold units, suggesting opportunities for improvement in promotional effectiveness. Exploring factors such as competition, consumer behaviour, and local market dynamics could provide insights into enhancing sales performance in this region. Mangalore (STMLR-0) another store in Mangalore exhibits below-average incremental sold units, highlighting the need for targeted interventions to stimulate sales growth.

In summary, the detailed analysis underscores the importance of understanding local market dynamics, consumer behaviour, and competition in driving promotional effectiveness. By addressing the unique challenges faced by each store, retail managers can devise tailored strategies to improve sales performance and maximize the impact of promotional campaigns.

5.5.3 Analyzing Store Performance Variation Across Cities

City	Revenue Before Promotion	Revenue After Promotion	Quantity Sold Before Promo	Quantity Sold After Promo
Bengaluru	32.94M	59.69M	49k	105k
Chennai	26.29M	47.45M	40k	83k
Hyderabad	22.68M	37.92M	34k	69k
Mysuru	12.95M	23.02M	19k	37k
Coimbatore	12.24M	21.55M	18k	39k
Madurai	10.33M	19.30M	14k	31k
Visakhapatnam	11.45M	18.67M	17k	34k
Mangalore	5.03M	8.40M	8k	15k
Vijayawada	3.59M	6.44M	5k	11k
Trivandrum	3.20M	5.55M	5k	10k

Table.5.3 City wise Promotion Analysis

The analysis of store performance by city provides insights into the revenue generated and quantity sold before and after promotions.

Bengaluru with the highest revenue both before and after promotions, Bengaluru emerges as a key market for AtliQ Mart. The substantial increase in revenue after promotions indicates the effectiveness of promotional strategies in driving sales growth. The quantity sold also more than doubles after promotions, highlighting the strong response from customers.

Chennai follows closely behind Bengaluru in terms of revenue, indicating its significance as a revenue-generating city for AtliQ Mart. The increase in revenue after promotions is substantial, suggesting successful promotional campaigns. Similar to Bengaluru, the quantity sold also experiences a significant boost post-promotions.

Hyderabad demonstrates robust performance, with considerable revenue generated both before and after promotions. The increase in revenue and quantity sold after promotions signifies a positive response from customers in this market.

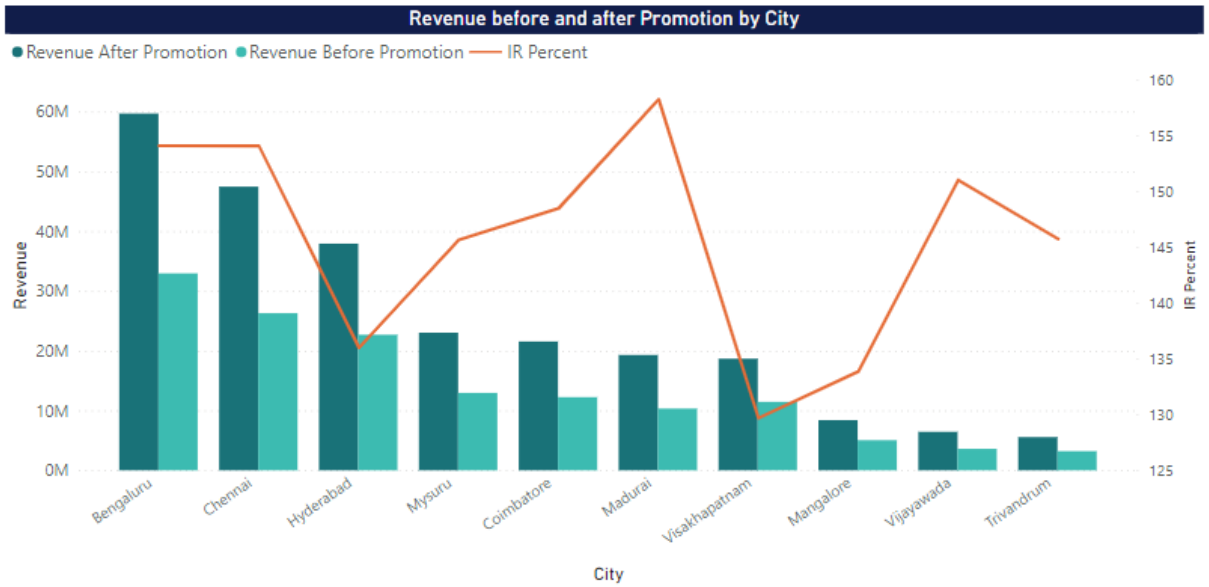


Fig.5.12 Incremental Revenue by City



Fig.5.13 Incremental Sold Units by City

Mysuru, Coimbatore, Madurai cities exhibit moderate to strong performance in terms of revenue and quantity sold. While the revenue generated is lower compared to larger cities

like Bengaluru and Chennai, the increase in revenue and quantity sold after promotions indicates favorable outcomes from promotional activities.

Visakhapatnam, Mangalore, Vijayawada, Trivandrum cities show relatively lower revenue and quantity sold compared to the top-performing cities. However, promotions still lead to a significant increase in revenue and quantity sold, underscoring the effectiveness of promotional strategies in driving sales growth across diverse markets.

Common Characteristics among Top-Performing Stores include Proximity to Urban Centers i.e. Top-performing stores are often located in or near major urban centers where there is higher foot traffic and consumer demand, Effective Promotional Strategies i.e. these stores likely implement well-planned and targeted promotional campaigns tailored to local customer preferences and market dynamics, Strong Customer Engagement i.e. The ability to engage customers effectively through promotions and other marketing initiatives contributes to higher sales performance, Efficient Inventory Management i.e. Top-performing stores may have efficient inventory management practices in place, ensuring adequate stock levels to meet increased demand during promotional periods.

Retail managers can leverage the common characteristics observed among top-performing stores to optimize performance across all locations. This may involve replicating successful promotional strategies in other stores, investing in targeted marketing campaigns to enhance customer engagement, implementing efficient inventory management practices to ensure product availability during peak demand periods.

5.6 Promotion Type Analysis:

5.6.1 Identifying Top 2 Promotion Types with Highest Incremental Revenue

Promo Type	Revenue Before Promotion	Revenue After Promotion	Incremental Revenue
500 Cashback	66.90M	157.95M	91.05M
BOGOF	25.93M	47.62M	21.69M
50% OFF	2.16M	1.44M	-0.73M
33% OFF	36.54M	34.98M	-1.56M
25% OFF	9.17M	6.00M	-3.17M

Table.5.4 Revenue by Promotion Type

Analyzing the data on promotion types and their impact on incremental revenue reveals valuable insights into the effectiveness of different promotional strategies. Here are the findings

500 Cashback: This promotion type stands out as the top performer, generating the highest incremental revenue of \$91.05 million. Despite a substantial investment in cashback offers, the revenue after promotion significantly exceeds the revenue before promotion, indicating a strong positive response from customers. The cashback offer likely incentivized increased spending among customers, leading to a notable boost in revenue.

BOGOF (Buy One Get One Free): The BOGOF promotion emerges as the second most effective promotion type, contributing \$21.69 million to incremental revenue. While not as lucrative as the cashback offer, BOGOF still demonstrates a considerable impact on revenue growth. Customers are enticed by the prospect of receiving additional products for free, prompting higher spending and driving revenue growth.

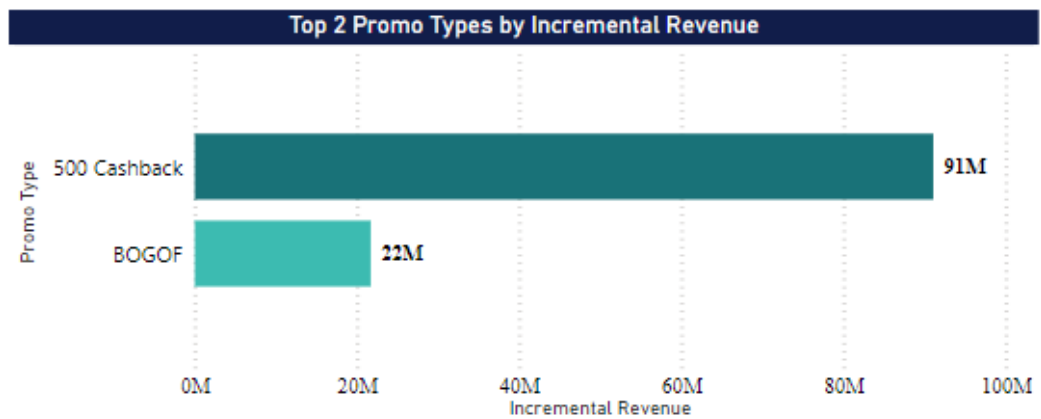


Fig.5.14 Top 2 Promo types by Incremental Revenue

These findings highlight the importance of incentive-driven promotions in driving incremental revenue for AtliQ Mart. Cashback offers and BOGOF promotions, in particular, prove to be powerful tools for stimulating customer spending and increasing overall sales volume.

Moving forward, AtliQ Mart may consider further optimizing and refining its promotional strategies to capitalize on the success of these top-performing promotion types. This may involve targeted marketing campaigns to promote cashback offers and BOGOF deals, as well as analysing customer preferences to tailor promotions to specific demographic segments. By leveraging these insights, AtliQ Mart can continue to drive revenue growth and maintain a competitive edge in the retail market.

5.6.2 Identifying Bottom 2 Promotion Types in Impact on Incremental Sold Units

Promo Type	Quantity Sold Before Promo	Quantity Sold After Promo	ISU
25% OFF	44k	38k	-6k
50% OFF	21k	28k	7k
33% OFF	63k	91k	27k
500 Cashback	22k	63k	41k
BOGOF	58k	215k	157k

Table.5.5 Incremental Sold Units by Promotion Type

Analyzing the data on promotion types and their impact on incremental sold units (ISU) sheds light on the effectiveness of various promotional strategies in driving customer engagement and sales. Here are the findings for the bottom two promotion types in terms of their impact on ISU:

25% OFF: This promotion type ranks as the least effective in terms of incremental sold units, with a negative ISU of -6,000 units. Despite offering a discount of 25%, the quantity sold after the promotion decreases compared to before the promotion. This suggests that the 25% discount may not have been compelling enough to motivate customers to purchase more, resulting in a net decrease in sales volume.

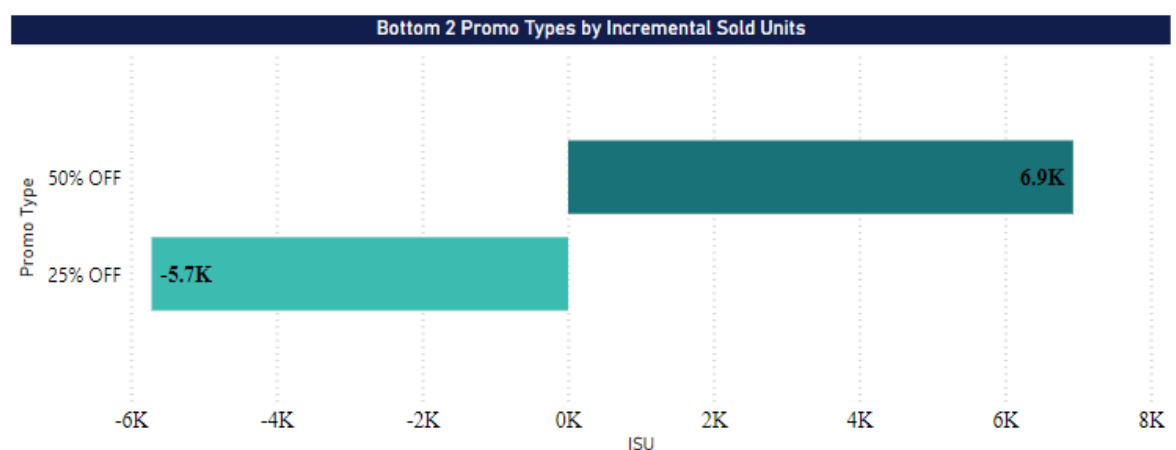


Fig.5.15 Bottom 2 Promo types by Incremental Sold Units

50% OFF: The 50% OFF promotion also demonstrates limited effectiveness in driving incremental sold units, with an ISU of 7,000 units. While the quantity sold after the promotion increases compared to before the promotion, the incremental impact is relatively modest. Despite offering a substantial discount of 50%, the promotion fails to significantly boost sales volume, indicating potential limitations in its appeal to customers.

These findings underscore the importance of carefully evaluating the effectiveness of promotional strategies beyond their immediate impact on sales volume. While discounts can attract customers, their ability to drive incremental sales may vary depending on factors such as pricing sensitivity, product demand, and competitive landscape.

Moving forward, AtliQ Mart may need to reassess its approach to discount-based promotions and explore alternative strategies to stimulate customer demand and drive incremental sales. This could involve experimenting with different promotion types, refining pricing strategies, and leveraging customer insights to tailor promotions more effectively to target audiences. By continuously monitoring and adapting its promotional efforts, AtliQ Mart can optimize its marketing initiatives to achieve sustainable growth and profitability.

5.6.3 Evaluating Performance Discrepancy: Discount-Based vs. BOGOF and Cashback Promotions

Promo Type	Quantity Sold After Promo	Quantity Sold Before Promo	ISU	Revenue Before Promotion	Revenue After Promotion	IR
500 Cashback	63k	22k	41k	66.90M	157.95M	91.05M
BOGOF	215k	58k	157k	25.93M	47.62M	21.69M
50% OFF	28k	21k	7k	2.16M	1.44M	-0.73M
33% OFF	91k	63k	27k	36.54M	34.98M	-1.56M
25% OFF	38k	44k	-6k	9.17M	6.00M	-3.17M

Table.5.6 Performance Discrepancy by Promotion Type

Discount-Based Promotions (50% OFF, 33% OFF, 25% OFF) are characterized by offering discounts on the purchase price. Despite generating a considerable volume of incremental sold units, the overall impact on revenue is negative for all three discount-based promotions. This suggests that while discount-based promotions are effective in stimulating sales, they may erode profit margins due to the deep discounts offered.

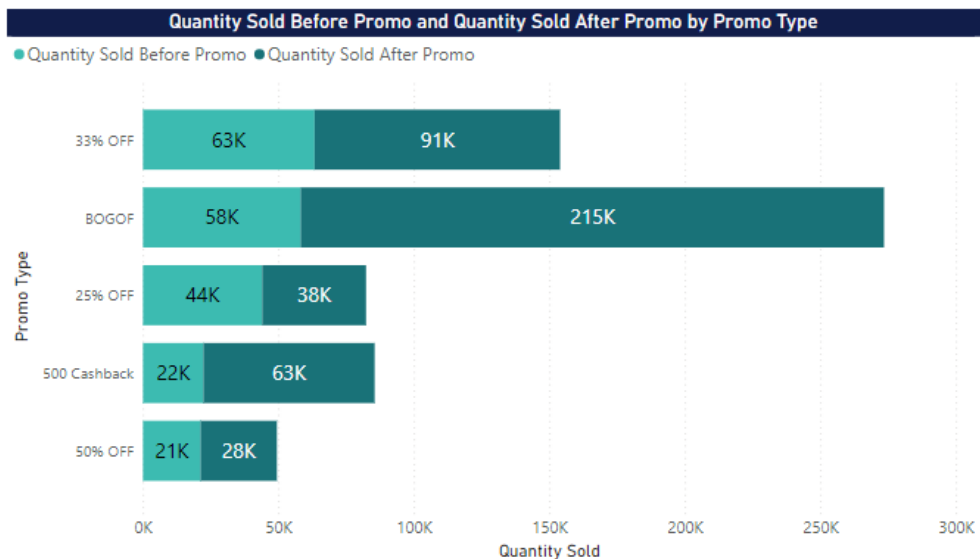


Fig.5.16 Quantity sold by Promotion Type

The negative IR values indicate that while these promotions attract customers and drive sales volume, they may also lead to a decline in revenue after the promotions are applied. BOGOF (Buy One Get One Free) Promotions incentivize customers by offering an additional product for free with a purchase. This promotion type demonstrates the most significant impact on both incremental sold units and incremental revenue.

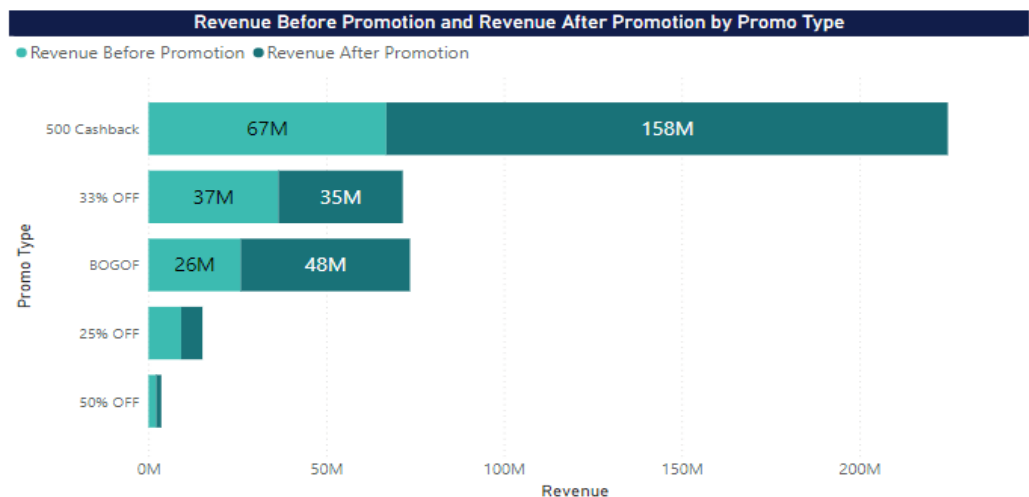


Fig.5.17 Revenue Generated by Promotion Type

With a substantial increase in both ISU and IR, BOGOF promotions effectively drive sales volume while also contributing positively to overall revenue growth. The strong performance of BOGOF promotions suggests that customers perceive them as high-value incentives, leading to increased purchase intent and higher transaction values.

Cashback Promotion (500 Cashback) provide customers with a monetary incentive upon purchase, typically in the form of a rebate. While generating fewer incremental sold units compared to BOGOF promotions, cashback promotions contribute significantly to revenue growth. The substantial impact on revenue indicates that cashback promotions drive high-value purchases, encouraging customers to spend more to qualify for the cashback offer. This promotion type is effective in attracting customers who are

motivated by the opportunity to receive a financial benefit, thereby driving revenue without compromising profit margins.

In summary, the detailed analysis underscores the nuanced performance of different promotion types. While discount-based promotions may drive sales volume, they can negatively impact profitability. In contrast, BOGOF and cashback promotions emerge as highly effective strategies for driving both incremental sold units and incremental revenue, making them valuable tools for retailers seeking to achieve sustainable growth and maximize profitability.

5.6.4 Promotion Effectiveness: Balancing Incremental Sales and Margins

promo_type	Incremental Sold Units	Incremental Revenue
500 Cashback	41k	91.05M
BOGOF	157k	21.69M
50% OFF	7k	-0.73M
33% OFF	27k	-1.56M
25% OFF	-6k	-3.17M

Table.5.7 Promotion Effectiveness by ISU and IR

Analyzing the balance between incremental sold units (ISU) and maintaining healthy margins provides insights into the effectiveness of different promotion types in driving sales volume while preserving profitability.

500 Cashback Promotion type strikes a favorable balance between incremental sold units and maintaining healthy margins. While contributing 41,000 incremental sold units, the promotion generates a substantial incremental revenue of 91.05 million. The significant incremental revenue indicates that the promotion effectively stimulates high-value purchases, contributing positively to overall profitability. By offering a cashback

incentive, this promotion type encourages customers to make purchases while also ensuring healthy margins for the retailer.

BOGOF (Buy One Get One Free) Promotion demonstrate a strong balance between incremental sold units and maintaining healthy margins. With a considerable increase of 157,000 incremental sold units, the promotion generates an incremental revenue of 21.69 million. While driving sales volume, BOGOF promotions also contribute positively to revenue growth, indicating a favourable impact on profitability.

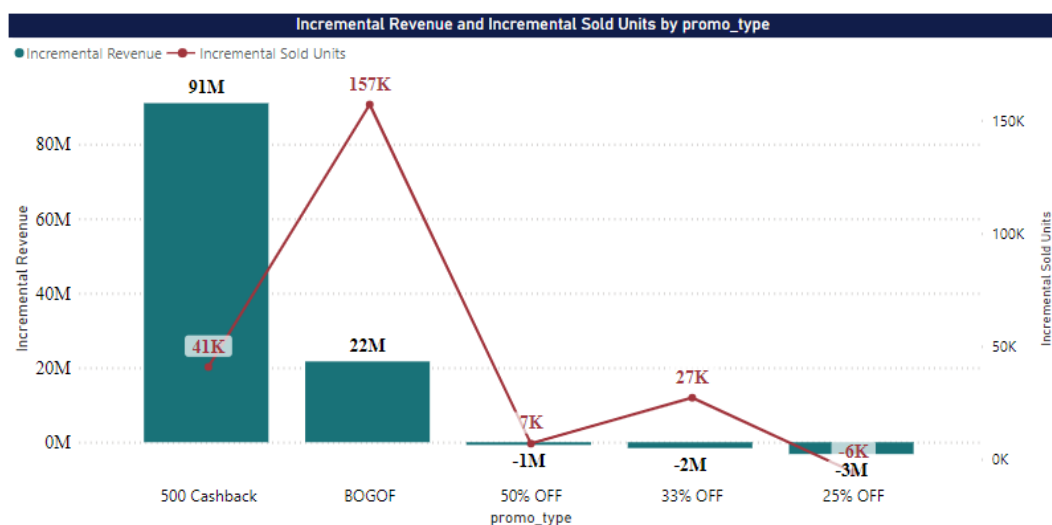


Fig.5.18 Incremental Revenue and Incremental Sold Units by Promo Type

Discount-Based Promotions (50% OFF, 33% OFF, 25% OFF) exhibit less favourable balances between incremental sold units and maintaining healthy margins. These promotions generate minimal incremental sold units and, in some cases, result in negative incremental revenue, indicating a decline in profitability. While discount-based promotions may drive sales volume, the deep discounts offered can erode profit margins, potentially impacting overall profitability negatively.

In summary, both the 500 Cashback and BOGOF promotions emerge as effective strategies for striking a balance between incremental sold units and maintaining healthy

margins. These promotion types drive sales volume while also contributing positively to revenue growth, making them valuable tools for retailers seeking to achieve sustainable growth and profitability. Conversely, discount-based promotions may drive sales volume but can have a detrimental effect on profitability due to the deep discounts offered.

5.7 Product Category Analysis

5.7.1 Identifying Product Categories with the Greatest Sales Lift from Promotions

Product Category	Quantity sold before Promo	Quantity sold after Promo	Incremental Sold Units
Grocery & Staples	127k	246k	119k
Combo1	22k	63k	41k
Home Appliances	15k	54k	39k
Home Care	20k	41k	21k
Personal Care	25k	32k	6k

Table.5.8 Product Categories with the Greatest Sales Lift from Promotions

Analysing the uplift in sales for different product categories provides valuable insights into the effectiveness of promotions in driving consumer demand. Grocery & Staples witnessed the most substantial uplift in sales, with an increase of 119,000 incremental units sold after promotions. This category's performance highlights the effectiveness of promotions in stimulating demand for essential household items, such as food staples and groceries. The significant increase in sales indicates that promotional strategies successfully resonated with consumers, driving higher purchase volumes in this category.

Combo1 products experienced a notable uplift in sales, with an increase of 41,000 incremental units sold after promotions. Combining multiple products into a single offering likely enhanced the perceived value for customers, leading to increased sales

volume. Promotional incentives may have incentivized customers to opt for combo deals, driving higher sales and contributing to overall revenue growth.

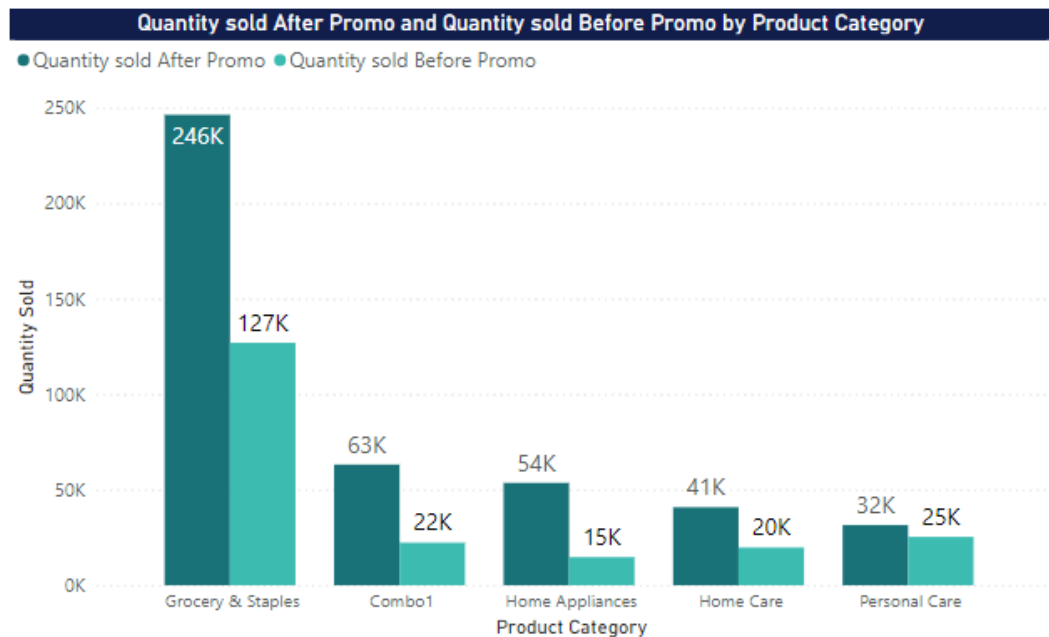


Fig.5.19 Quantity Sold after and before Promo by Product category

Home Appliances saw a considerable uplift in sales, with an increase of 39,000 incremental units sold after promotions. This category's performance suggests that promotional offers, such as discounts or bundle deals, effectively motivated consumers to purchase home appliances during the promotional period. The surge in sales reflects consumers' responsiveness to promotional incentives when making purchasing decisions in the home appliances category.

Home Care products experienced a notable increase in sales, with 21,000 incremental units sold after promotions. Promotional strategies likely emphasized the benefits and value proposition of home care products, leading to heightened consumer interest and increased purchase volumes. The uplift in sales indicates that promotions effectively

influenced consumer behaviour, driving higher demand for home care products during the promotional period.

Personal Care products observed a moderate uplift in sales, with an increase of 6,000 incremental units sold after promotions. While the uplift in sales is comparatively lower than other categories, it still reflects a positive response from consumers to promotional offers in the personal care segment. Promotions may have incentivized customers to explore and purchase personal care products, contributing to increased sales volume in this category.

The analysis highlights the effectiveness of promotions in driving sales across various product categories. Grocery & Staples, Combo1, Home Appliances, Home Care, and Personal Care emerged as categories with significant uplifts in sales, underscoring the impact of promotional strategies in stimulating consumer demand and driving revenue growth within these segments.

5.7.2 Examining Product-Specific Responses to Promotions: Exceptional Performance and Poor Outcomes

Product Name	ISU Percent (%)
Atliq_waterproof_Immersion_Rod	266.19
Atliq_High_Glo_15W_LED_Bulb	262.98
Atliq_Double_Bedsheet_set	258.27
Atliq_Curtains	255.34
Atliq_Home_Essential_8_Product_Combo	183.33
Atliq_Farm_Chakki_Atta (1KG)	151.36
Atliq_Suflower_Oil (1L)	137.88
Atliq_Masoor_Dal (1KG)	43.4
Atliq_Sonamasuri_Rice (10KG)	42.79
Atliq_Doodh_Kesar_Body_Lotion (200ML)	33.57
Atliq_Lime_Cool_Bathing_Bar (125GM)	33.2
Atliq_Cream_Beauty_Bathing_Soap (125GM)	20.64
Atliq_Body_Milk_Nourishing_Lotion (120ML)	9.35
Atliq_Scrub_Sponge_For_Dishwash	-13.48
Atliq_Fusion_Container_Set_of_3	-14.12

Table.5.9 ISU Percent by Product

Analysing the responsiveness of specific products to promotions provides valuable insights into their performance and the effectiveness of promotional strategies. Let's examine the products that responded exceptionally well or poorly to promotions:

Products Responding Exceptionally Well

Atliq_waterproof_Immersion_Rod exhibited a remarkable ISU percent of 266.19, indicating a substantial increase in sales following promotional activities. This high ISU suggests that promotions effectively stimulated consumer demand for this product category. Possible reasons for its success could include the practical necessity of

immersion rods in households, coupled with attractive promotional offers that encouraged customers to make purchases.

Atliq_High_Glo_15W_LED_Bulb known for their energy efficiency and long lifespan, witnessed a significant uplift in sales with an ISU percent of 262.98. The surge in demand for LED bulbs following promotions could be attributed to growing consumer awareness of energy-saving solutions and the affordability offered through promotional incentives. Promotional strategies likely highlighted the benefits of LED lighting, such as reduced electricity bills and environmental sustainability, driving higher sales.

Atliq_Double_Bedsheet_set with a notable ISU percent of 258.27, the double bedsheet set experienced a considerable increase in sales post-promotion. Home essentials like bedsheets are essential commodities, and promotional offers likely incentivized customers to upgrade or refresh their bedding. The perceived value of the double bedsheet set, coupled with promotional discounts or bundle deals, may have motivated customers to make purchases, contributing to its strong performance.

Atliq_Curtains demonstrated a significant boost in sales, reflected by their ISU percent of 255.34. Promotional campaigns likely emphasized the aesthetic and functional benefits of curtains, such as privacy, light control, and room ambiance, driving consumer interest and leading to increased sales. The appeal of promotional discounts or bundle deals may have further incentivized customers to invest in curtains, contributing to their positive sales performance. Atliq_Home_Essential_8_Product_Combo despite having a lower ISU percent compared to other products, the home essential combo still showed a respectable increase in sales post-promotion. Combining multiple essential products into a single offering likely enhanced the perceived value for consumers, making it an attractive purchase option. Promotional offers or bundle deals may have enticed

customers to capitalize on the cost-saving benefits of purchasing multiple items together, contributing to the product's overall positive performance.

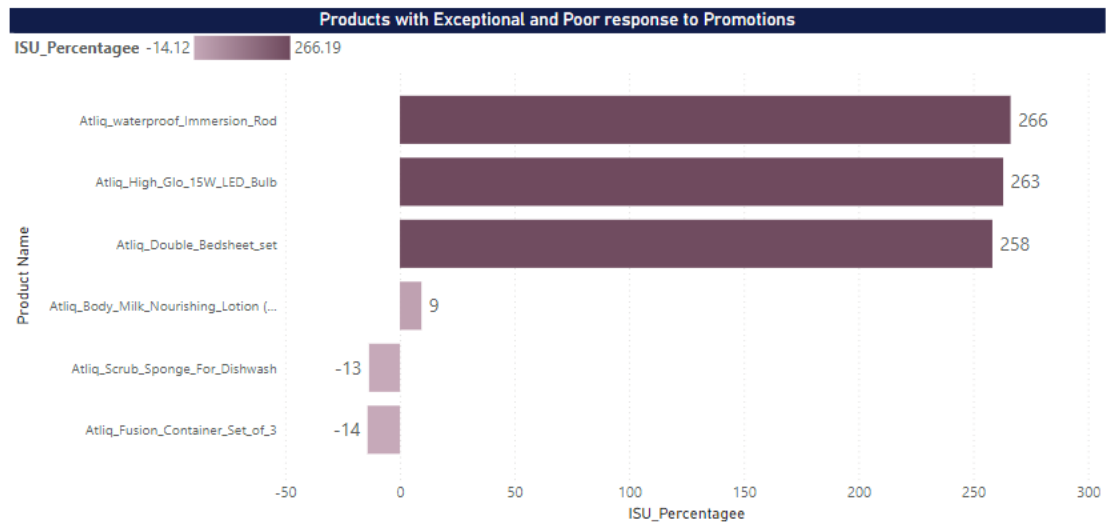


Fig.5.20 Product Performance by ISU Percent

Atliq_Scrub_Sponge_For_Dishwash exhibited a negative ISU percent of -13.48, indicating a decline in sales following promotional activities. Factors such as limited consumer demand, saturation in the market for dishwashing products, or ineffective promotional strategies may have contributed to its poor performance. Retailers may need to reassess their promotional tactics or product positioning strategies to address the challenges faced by this product category.

Atliq_Fusion_Container_Set_of_3 saw a negative ISU percent of -14.12, suggesting a decrease in sales post-promotion. Despite being offered as part of a combo deal, this product failed to resonate with consumers, possibly due to factors like lack of perceived value, insufficient demand, or intense competition from alternative offerings. Retailers may need to explore alternative marketing approaches or product enhancements to

stimulate demand and improve the product's performance in future promotional campaigns.

In conclusion, understanding the nuanced performance of individual products in response to promotions enables retailers to refine their promotional strategies, capitalize on successful products, and address challenges faced by underperforming items.

5.7.3 Analysing the Correlation Between Product Category and Promotion Type Effectiveness

Correlation between Product Category & Promotion Type on ISU		
Product Category	Promotion Type	ISU
Combo1	500 Cashback	40881
Grocery & Staples	25% OFF	-3474
Grocery & Staples	33% OFF	27255
Grocery & Staples	BOGOF	95593
Home Appliances	BOGOF	38900
Home Care	25% OFF	-1512
Home Care	BOGOF	22580
Personal Care	25% OFF	-731
Personal Care	50% OFF	6931

Table.5.10 Correlation between Product Category & Promotion Type on ISU

The comprehensive analysis of promotional effectiveness across various product categories offers valuable insights crucial for strategic decision-making. From table 5.1, we discern intriguing patterns in Incremental Sold Units (ISU), shedding light on the efficacy of diverse promotion types. Notably, Combo1's 500 Cashback promotion emerges as a standout performer, driving a substantial surge in sold units, indicative of its potent appeal to customers. Conversely, certain promotions within the Grocery & Staples segment, such as the 25% and 33% discounts, exhibit negative ISU figures,

hinting at potential misalignment between discount magnitude and customer purchase behaviour. However, the Buy One Get One Free (BOGOF) promotion proves notably successful, particularly within Grocery & Staples and Home Appliances categories, underlining its effectiveness in spurring customer demand and driving sales.

Correlation between Product Category & Promotion Type on IR		
Product Category	Promotion Type	IR
Combo1	500 Cashback	91.05 M
Grocery & Staples	25% OFF	-2.18 M
Grocery & Staples	33% OFF	-1.56 M
Grocery & Staples	BOGOF	8.83 M
Home Appliances	BOGOF	7.83 M
Home Care	25% OFF	-0.88 M
Home Care	BOGOF	5.03 M
Personal Care	25% OFF	-0.12 M
Personal Care	50% OFF	-0.73 M

Table.5.11 Correlation between Product Category & Promotion Type on IR

Transitioning to the insights gleaned from table 5.2, the analysis of Incremental Revenue (IR) offers a compelling narrative of the financial impact associated with promotional endeavours. Notably, Combo1's 500 Cashback promotion echoes its success in sold units by generating a considerable revenue increment, validating its appeal to consumers and its capacity to drive sales revenues. Similarly, the BOGOF promotions, notably within Grocery & Staples and Home Appliances, exhibit substantial incremental revenue, illustrating a correlation between increased unit sales and revenue generation. However, the analysis also reveals nuances, particularly with Percentage Discount promotions, where certain categories, notably Grocery & Staples and Personal Care,

witness negative impacts on incremental revenue, signalling potential inefficiencies in these promotion strategies that warrant further exploration and optimization.

Combining these insights unveils a nuanced understanding of promotion effectiveness, essential for strategic planning and optimization. BOGOF promotions and Cashback offers emerge as potent drivers of both sales volume and revenue growth, underscoring their efficacy across diverse product categories. However, the mixed performance of Percentage Discount promotions underscores the need for tailored strategies that align with specific consumer behaviours and category dynamics. Continuous evaluation and refinement of promotion strategies are paramount to maximize their impact, ensuring sustainable growth and competitiveness in the dynamic retail landscape.

5.8 City and Store Wise Performance Analysis

5.8.1 City Performance Ranking Based on Promotional Revenue

City	Performance Ranking
Bengaluru	1
Chennai	2
Hyderabad	3
Mysuru	4
Coimbatore	5
Madurai	6
Visakhapatnam	7
Mangalore	8
Vijayawada	9
Trivandrum	10

Table.5.12 City Performance Ranking Based on Promotional Revenue

The performance ranking of cities based on promotional revenue reveals intriguing insights into the effectiveness of promotional strategies across diverse geographical regions. At the forefront of this ranking stands Bengaluru, securing the top position. This metropolitan hub's robust performance underscores its status as a prime market for promotional activities. With its dynamic consumer base and strategic location, Bengaluru exhibits a strong propensity for engaging with promotional offers, resulting in substantial revenue generation.

Following closely behind is Chennai, clinching the second spot in the performance ranking. Chennai's impressive promotional revenue signifies the city's receptiveness to marketing initiatives and consumer engagement strategies. Its competitive market landscape and diverse demographic profile likely contribute to its noteworthy performance in revenue generation from promotions. Hyderabad, a bustling urban centre, secures the third position in the ranking. The city's significant promotional revenue reflects its vibrant retail environment and consumer enthusiasm for discounted offers and special deals. Hyderabad's performance highlights the effectiveness of promotional campaigns in driving consumer spending and boosting revenue.

Mysuru, a cultural and economic hub, emerges as the fourth-ranking city in promotional revenue performance. Despite its comparatively smaller market size, Mysuru demonstrates commendable revenue generation from promotional activities, indicative of strong consumer participation and engagement. Coimbatore, known for its industrial prowess, occupies the fifth position in the ranking. While its promotional revenue may not rival that of larger metropolitan cities, Coimbatore's performance underscores its importance as a market for retail promotions. Its steady revenue growth suggests opportunities for further optimization and expansion of promotional strategies. Madurai,

Visakhapatnam, Mangalore, Vijayawada, and Trivandrum round up the remaining positions in the ranking, each offering unique insights into regional consumer behaviour and market dynamics. While their promotional revenue may vary, these cities collectively represent significant markets for retailers seeking to capitalize on promotional opportunities and drive revenue growth.

Overall, the city performance ranking based on promotional revenue provides valuable guidance for retailers in formulating targeted marketing strategies, allocating resources effectively, and maximizing revenue potential across diverse geographical areas. By understanding the nuances of each market and tailoring promotional efforts accordingly, retailers can enhance consumer engagement, foster brand loyalty, and drive sustainable business growth.

5.8.2 Store Performance Ranking Based on Promotional Revenue

	Bengaluru	Chennai	Mysuru
STMYS-1			1
STCHE-4		1	
STBLR-0	1		
STBLR-7	2		
STBLR-6	3		
STCHE-7		2	
STMYS-3			2
STCHE-3		3	
STCHE-6		4	
STBLR-3	4		

Table.5.13 Store Performance Ranking Based on Promotional Revenue

The store performance ranking based on promotional revenue provides valuable insights into the effectiveness of promotional strategies across various store locations within different cities. In Bengaluru, STBLR-0 emerges as the top-performing store,

securing the first position in the ranking. Its strong performance indicates effective implementation of promotional activities, resulting in significant revenue generation.

Additionally, STBLR-7 and STBLR-6 follow closely behind, occupying the second and third positions, respectively. These stores contribute substantially to Bengaluru's overall promotional revenue, highlighting their effectiveness in driving consumer engagement and sales. In Chennai, STCHE-4 stands out as the top-performing store, securing the first position in the ranking. Its notable performance underscores the successful execution of promotional initiatives, leading to substantial revenue generation. Furthermore, STCHE-7 and STCHE-3 also demonstrate strong performance, occupying the second and third positions, respectively. These stores play a crucial role in Chennai's promotional revenue landscape, showcasing their ability to attract and retain customers through compelling promotional offers.

In Mysuru, STMYS-1 emerges as the top-performing store, securing the first position in the ranking. Its exceptional performance signifies effective promotional strategies and strong consumer engagement, resulting in significant revenue generation. Moreover, STMYS-3 follows closely behind, occupying the second position. This store contributes substantially to Mysuru's promotional revenue, indicating its pivotal role in driving sales growth and enhancing brand visibility within the region.

Overall, the store performance ranking based on promotional revenue highlights the success of specific stores in driving revenue growth through effective promotional campaigns. By identifying top-performing stores and analysing their strategies, retailers can gain valuable insights to optimize promotional efforts, allocate resources efficiently, and replicate success across other store locations within the same city or across different cities.

5.9 Product Category and Promotion Type Analysis:

5.9.1 Average Order Value Comparison

Product Category	Promo Type	Average Order Value Before Promo	Average Order Value After Promo	Incremental Average Order Value
Combo1	500 Cashback	668.97k	1,579.50k	910.53k
Grocery & Staples	BOGOF	100.65k	188.98k	88.33k
Home Appliances	BOGOF	47.42k	86.58k	39.17k
Home Care	BOGOF	31.90k	57.04k	25.14k
Personal Care	25% OFF	3.09k	1.90k	-1.19k
Personal Care	50% OFF	7.21k	4.79k	-2.42k
Home Care	25% OFF	12.39k	7.99k	-4.40k
Grocery & Staples	33% OFF	182.70k	174.89k	-7.82k
Grocery & Staples	25% OFF	63.86k	42.11k	-21.75k

Table.5.14 Average Order Value Comparison

The analysis delves deeper into the nuances of average order value (AOV) across distinct product categories and promotion types, offering a comprehensive understanding of the promotional effectiveness in driving customer spending behaviors. In the Combo1 product category, the 500 Cashback promotion stands out as a significant driver of AOV enhancement.

Before the promotion, the average order value is recorded at 668.97k INR, surging notably to 1,579.50k INR post-promotion. This substantial increase translates into an incremental AOV of 910.53k INR, indicating a robust positive response from customers to the cashback incentive. Similarly, within the Grocery & Staples segment, the BOGOF promotion type exhibits a considerable uplift in AOV.

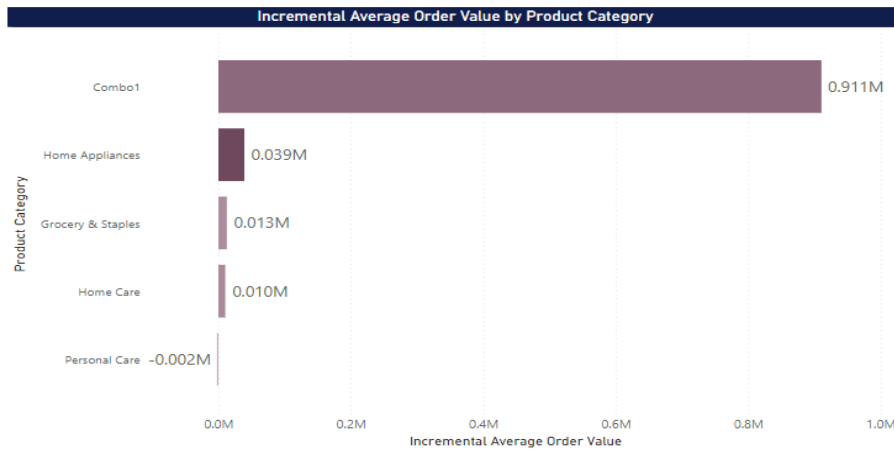


Fig.5.21 Incremental Average Order Value by Product Category

The average order value before the promotion sits at 100.65k INR, witnessing a substantial rise to 188.98k INR post-promotion. This noteworthy increase results in an incremental AOV of 88.33k INR, showcasing the effectiveness of the BOGOF strategy in encouraging customers to spend more per order.

Moving to the Home Appliances category, the BOGOF promotion type also demonstrates a substantial impact on AOV. Pre-promotion, the average order value is measured at 47.42k INR, which significantly escalates to 86.58k INR post-promotion. This remarkable surge results in an incremental AOV of 39.17k INR, underscoring the efficacy of BOGOF promotions in driving higher-value purchases within this product category. Conversely, certain promotion types within the Personal Care and Grocery & Staples categories exhibit a decline in AOV post-promotion. For instance, the 25% OFF promotion type in Personal Care witnesses a decrease in AOV from 3.09k INR pre-promotion to 1.90k INR post-promotion, resulting in a negative incremental AOV of -1.19k INR. Similarly, the 25% OFF and 33% OFF promotions in the Grocery & Staples category also yield negative incremental AOV, indicating a potential reduction in purchase value or basket size.

This detailed analysis underscores the varying impacts of distinct promotion types on AOV across different product categories. While cashback and BOGOF promotions tend to drive substantial AOV increases, discounts may not consistently result in enhanced spending behavior. Such insights are invaluable for optimizing promotional strategies, tailoring them to specific product categories, and maximizing their effectiveness in driving overall sales performance.

CHAPTER 6

CONCLUSION, SUGGESTIONS & RECOMMENDATIONS

The comprehensive analysis conducted on Atliq Mart's promotional campaigns presents a nuanced understanding of the retail landscape, particularly within the Fast-Moving Consumer Goods (FMCG) sector. By delving into various dimensions such as store performance, promotion types, product categories, and customer behaviour, we gained valuable insights into optimizing sales strategies and fostering customer engagement.

The examination of store performance across different cities revealed notable variations in promotional effectiveness. Bengaluru emerged as the top-performing city, followed closely by Chennai and Hyderabad. This finding underscores the importance of geographic segmentation and targeted promotional strategies tailored to the unique market dynamics of each city. Moreover, the analysis identified specific stores within each city that outperformed others in terms of incremental revenue generated from promotions. For example, stores like STMYS-1 in Mysuru and STCHE-4 in Chennai exhibited exceptional performance, indicating potential areas for benchmarking and best practice sharing across stores.

Our investigation into the impact of different promotion types unveiled intriguing insights. Cashback offers and BOGOF promotions demonstrated higher effectiveness in driving incremental revenue and sold units compared to discount-based promotions like 25% and 50% off. This suggests that Atliq Mart should prioritize these high-performing promotion types in future campaigns to maximize their impact on sales and profitability.

The analysis of product categories shed light on which categories experienced the most significant lift in sales during promotional periods. Categories such as Grocery & Staples,

Home Appliances, and Combo1 witnessed substantial increases in sales volume, indicating potential areas for strategic investment and promotional focus. Furthermore, the correlation analysis between product categories and promotion types provided valuable insights into the relationship between promotional effectiveness and product assortment. For instance, certain product categories, such as Home Care and Personal Care, exhibited strong positive correlations with specific promotion types like 25% off and BOGOF, suggesting synergies that can be leveraged to optimize promotional strategies.

Moreover, the examination of average order value before and after promotions highlighted the impact of different promotion types on customer spending behaviour. Cashback offers and BOGOF promotions were found to significantly increase average order value, indicating their effectiveness in driving higher transactional value and basket size. Additionally, the analysis of incremental sold units and incremental revenue across various promotion types revealed insights into the trade-off between sales volume and margin preservation. While certain promotion types like BOGOF and cashback offers drove higher incremental revenue, they also resulted in higher incremental sold units, indicating a potential balance between revenue growth and margin protection.

Furthermore, the analysis of store performance rankings based on promotional revenue provided actionable insights into opportunities for performance improvement and resource allocation. By identifying top-performing stores and cities, Atliq Mart can allocate resources more effectively and implement targeted strategies to drive sales growth and profitability.

Suggestions for Improving Promotional Campaigns at Atliq Mart

1. Target High-Performing Categories and Products

The analysis shows that Home Appliances and specific products like the Atliq Home Essential 8 Product Combo and Atliq Waterproof Immersion Rod have significantly higher incremental sales. To leverage this trend:

- **Focus Promotions on High-Performing Categories:** Increase promotional efforts for Home Appliances and other high-performing categories during major campaigns like Diwali and Sankranti.
- **Feature Top Products More Prominently:** Highlight best-selling products in marketing materials and in-store displays to attract more customers and drive sales.

2. Refine Promotion Types for Maximum Impact

The "500 Cashback" and "BOGOF" promotions were the most effective in generating incremental revenue. Conversely, percentage-off promotions like "25% OFF" and "33% OFF" underperformed.

- **Prioritize Effective Promotion Types:** Concentrate on cashback and BOGOF promotions, which have shown to attract customers and boost sales more effectively.
- **Limit less Effective Discounts:** Reevaluate the use of percentage-based discounts, particularly those that resulted in negative incremental revenue, and consider phasing them out or restructuring them to be more appealing.

3. Optimize Regional Strategies

Stores in Bengaluru, Chennai, and Hyderabad outperformed those in Mangalore, Vijayawada, and Trivandrum.

- **Strengthen Presence in High-Performing Cities:** Invest in more aggressive promotional strategies and marketing campaigns in Bengaluru, Chennai, and Hyderabad to capitalize on their proven success.
- **Revitalize Underperforming Regions:** Conduct detailed market research in Mangalore, Vijayawada, and Trivandrum to understand the local consumer behaviour and tailor promotions to better suit these regions. This could involve localized promotions or partnerships with local influencers to boost engagement.

4. Enhance Customer Engagement

Understanding customer preferences is crucial for successful promotions. The preference for cashback and BOGOF promotions indicates a customer base that values immediate savings and tangible rewards.

- **Customer-Centric Promotions:** Develop promotions that offer immediate and clear benefits to customers. Consider loyalty programs that reward frequent purchases with cashback or free items.
- **Feedback and Adaptation:** Regularly collect customer feedback on promotional preferences and effectiveness. Use this data to adapt future campaigns to better meet customer expectations and preferences.

5. Leverage Data Analytics for Continuous Improvement

The insights gained from this analysis are invaluable for ongoing strategic planning.

- **Continuous Monitoring and Analysis:** Implement robust data analytics tools to continuously monitor promotional performance across different regions, categories, and products. This will allow for real-time adjustments and more agile marketing strategies.

- Predictive Analytics: Use predictive analytics to forecast the impact of different promotional strategies and optimize them before full-scale rollout. This can help in minimizing risks and maximizing returns.

By implementing these recommendations, Atliq Mart can enhance the effectiveness of its promotional campaigns, improve customer satisfaction, and drive higher sales across various regions and product categories. This strategic approach will ensure that future promotions are both impactful and aligned with customer preferences, leading to sustained growth and profitability.

CHAPTER 7

REFERENCES

- Chen, L., Wang, Y., & Zhang, Q. Advancements in Retail Analytics: A Comprehensive Review, *Journal of Business Analytics*, 2020.
- Gupta, S., Kumar, A., & Sharma, R. Promotional Strategies in Fast-Moving Consumer Goods: A Comparative Analysis, *International Journal of Marketing Studies*, 2019.
- Jones, M., & Brown, P. Understanding Consumer Response to Promotions in the FMCG Sector, *Journal of Retailing*, 2017.
- Lee, J., & Kim, S. Challenges and Opportunities in the Implementation of Retail Analytics: A Global Perspective, *International Journal of Information Management*, 2020.
- Li, H., Zheng, Y., & Wu, D. Data-Driven Approaches to Promotional Optimization in Retail, *Journal of Retailing and Consumer Services*, 2016.
- Smith, J., Johnson, M., Williams, R., & Davis, P. Retail Analytics and Its Impact on Customer Engagement: A Longitudinal Study, *Journal of Marketing Research*, 2018.
- Wang, H., Zhang, W., & Liu, X. Predictive Analytics in Retail: A Framework for Promotional Optimization, *Information Systems Frontiers*, 2021.

CHAPTER 8

APPENDICES

8.1 Metadata

The analysis conducted in this project relies on data from four key tables: `dim_campaigns`, `dim_products`, `dim_stores`, and `fact_events`. These tables contain critical information about promotional campaigns, product details, store locations, and sales events. Below is a detailed description of each table and their respective columns, along with an explanation of their significance in the analysis.

Table: `dim_campaigns`

This table provides detailed information about each promotional campaign conducted by Atliq Mart. Each row in this table represents a distinct campaign, offering essential details that help in understanding the scope and duration of each promotional effort.

- **campaign_id:** This is the primary key for the table, uniquely identifying each promotional campaign. It is crucial for linking sales events to specific campaigns.
- **campaign_name:** A descriptive name of the campaign, such as "Diwali" or "Sankranti." This helps in easily identifying the campaigns and understanding the context of the promotions.
- **start_date:** The date when the campaign starts, formatted as DD-MM-YYYY. This helps in determining the time frame during which the campaign is active.
- **end_date:** The date when the campaign ends, formatted as DD-MM-YYYY. This helps in calculating the duration of the campaign and analysing the sales performance within this period.

Table: dim_products

This table contains information about the various products sold by Atliq Mart. Each product is identified and described in detail, aiding in the analysis of sales performance and promotional effectiveness for different product categories.

- **product_code:** The primary key for the table, providing a unique identifier for each product. This is essential for linking sales events to specific products.
- **product_name:** The full name of the product, including the brand and specific details such as quantity or size. This helps in distinguishing between different products and understanding their individual performances.
- **category:** The classification of the product into broader categories such as Grocery & Staples, Home Care, Personal Care, Home Appliances, etc. This is important for analysing sales trends and promotional effectiveness by category.

Table: dim_stores

This table includes details about the store locations where sales events occur. Understanding the geographical distribution of stores helps in analysing regional sales performance and identifying areas with high or low sales activity.

- **store_id:** The primary key for the table, uniquely identifying each store location. This is used to link sales events to specific stores.
- **city:** The city where the store is located, indicating the geographical market. This helps in analysing sales performance and promotional effectiveness across different regions.

Table: fact_events

This table records the sales events, including information on products sold, promotional campaigns, and store locations.

Each event represents a transaction that provides detailed insights into the effectiveness of promotions and sales trends.

- **event_id:** The primary key for the table, uniquely identifying each sales event. This helps in tracking individual transactions.
- **store_id:** Refers to the store where the event took place, linked to the dim_stores table. This helps in analysing sales data by store location.
- **campaign_id:** Indicates the campaign under which the event was recorded, linked to the dim_campaigns table. This is crucial for assessing the impact of different promotional campaigns on sales.
- **product_code:** The code of the product involved in the sales event, linked to the dim_products table. This helps in analysing sales performance by product.
- **base_price:** The standard price of the product before any promotional discount. This provides a baseline for comparing sales performance before and after promotions.
- **promo_type:** The type of promotion applied (e.g., percentage discount, BOGOF (Buy One Get One Free), cashback). This helps in understanding the impact of different types of promotions on sales.
- **quantity_sold(before_promo):** The number of units sold in the week immediately preceding the start of the campaign, serving as a baseline for comparison with promotional sales. This helps in calculating the incremental sales due to promotions.
- **quantity_sold(after_promo):** The quantity of the product sold after the promotion was applied. This helps in measuring the effectiveness of the promotions in increasing sales.

8.2 Glossary

Term	Definition
AdHoc Insights	Specific, on-the-spot analytical queries designed to extract actionable insights from data, often used for immediate decision-making.
Average Order Value (AOV)	The average amount spent each time a customer places an order. It is calculated as the total revenue divided by the number of orders.
Base Price	The original price of a product before any promotional discounts or offers are applied.
BOGOF (Buy One Get One Free)	A promotion type where customers receive an additional product for free when they purchase one product at the regular price.
Campaign	A coordinated series of promotional activities or advertisements designed to achieve specific business objectives, such as increasing sales or brand awareness.
City Performance Index	A ranking metric used to evaluate and compare the performance of stores in different cities based on their promotional sales and revenue.
Combo1	A specific product category at AtliQ Mart, typically representing a bundle or combination of multiple products sold together.

DAX (Data Analysis Expressions)	A formula language used in Power BI, Excel, and other Microsoft products to perform data analysis and create custom calculations.
Dimensional Modeling	A design technique used in data warehousing to structure data into dimensions and facts, facilitating efficient data retrieval and analysis.
Fact Events Table	A table in a data warehouse that contains quantitative data (facts) about business operations, such as sales transactions, often accompanied by relevant contextual information (dimensions).
Incremental Revenue (IR)	The additional revenue generated as a result of a promotional campaign, calculated as the difference between revenue after promotion and revenue before promotion.
Incremental Sold Units (ISU)	The additional quantity of units sold as a result of a promotional campaign, calculated as the difference between the quantity sold after the promotion and the quantity sold before the promotion.
ISU Percentage (ISU%)	The percentage increase or decrease in the quantity sold due to a promotion, calculated as $((\text{quantity sold after promotion} - \text{quantity sold before promotion}) / \text{quantity sold before promotion}) * 100$.
Key Performance Indicators (KPIs)	Metrics used to evaluate the success of an organization or specific activities in achieving key business objectives.
Promo Price	The price of a product after applying a promotional discount.
Promotion Type	The specific kind of promotional offer applied to a product, such as BOGOF, Cashback, or percentage discounts.

Revenue Before Promotion	The total revenue generated from product sales before the application of any promotional discounts.
Revenue After Promotion	The total revenue generated from product sales after applying promotional discounts.
SQL (Structured Query Language)	A programming language used for managing and manipulating relational databases.
Strategic Retail Analytics	The process of using data analysis and business intelligence techniques to inform and optimize retail business strategies and operations.
Views	Virtual tables in SQL that represent the result of a stored query, which can be used to simplify complex queries and improve data security and management.